

AfyaSolar Intelligence Platform

Developer Technical Specification

Re-invention Plan, Engineering Specification, and Implementation Guide

Ubuntu AfyaLink Company Limited

Dar es Salaam, Tanzania

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Audience: Engineering team, product, and external technical reviewers

Repository: github.com/ubuntuafyalink/afyasolar

Website: www.ubuntuafyalink.co.tz

Part 1 — Executive Summary

1.1 What this document is

This document is the engineering source of truth for AfyaSolar Intelligence Platform, the digital infrastructure operated by Ubuntu AfyaLink to deliver climate-resilient primary healthcare facility energy services across Tanzania and, in time, the broader East African region. It is written for engineers who will implement the platform, for product managers who will sequence the build, and for external technical reviewers who will assess the platform against international standards. It supersedes informal architecture notes and earlier engineering drafts.

The document covers the strategic re-invention of the current platform (which is live at github.com/ubuntuafyalink/afyasolar and powers thirteen commissioned facilities and thirty-three completed energy efficiency assessments), the target architecture under the climate-data-substrate model, the data collection specifications, the three persona dashboards (Facility Manager, Ubuntu AfyaLink Administrator, Impact Investor), the artificial intelligence model architecture, the integration list with specific Application Programming Interface acquisition instructions, the international standards conformance map (localised to Tanzania), the Energy-as-a-Service pricing architecture, and the implementation roadmap.

1.2 The strategic shift this document represents

The existing AfyaSolar platform is an energy services platform with a software side. The target platform is a climate-health data infrastructure with an energy services side. This is a meaningful repositioning. The existing platform's value to a facility comes from the energy hardware that is installed; the target platform's value to a facility comes from the data, the intelligence layer, and the operational support that surrounds the hardware. The hardware is the consequence of the platform decision, not the platform decision itself.

Three specific strategic shifts shape every subsequent technical choice in this document.

First, the platform abandons hardware-heavy telemetry. The original architectural plan envisioned smart energy meters, Modbus integration with inverters, cold-chain temperature loggers, and edge gateways at each facility. This added per-facility hardware cost of several hundred United States dollars, locked the platform into vendor cloud Application Programming Interfaces, and created an operational maintenance burden that does not scale beyond a few hundred facilities. The new approach uses climate data as the substrate, periodic manual inputs from Energy Champions as the truth, and artificial intelligence inference as the bridge. The marginal data acquisition cost for a new facility drops from several hundred dollars to zero.

Second, the platform is positioned as sovereign infrastructure rather than as a service. Following the Amini model, the platform is the substrate other actors build on top of. Ubuntu AfyaLink is the reference operator that proves the substrate works. Ministries, faith-based networks, Non-Governmental Organisations, Development Finance Institutions, academic researchers, parametric insurance underwriters, and carbon-market originators are all addressable stakeholders. Each consumes the platform through a different lens; all rest on the same data.

Third, the platform commits to the standards conformance trail. Every component is mapped to its relevant international standard: American Society of Heating, Refrigerating and Air-Conditioning

Engineers Standard 211 for audits, International Organization for Standardization 50001 for energy management, International Performance Measurement and Verification Protocol for savings verification, World Health Organization Guidance for climate-resilient health facilities, Intergovernmental Panel on Climate Change Sixth Assessment Report for climate science, International Organization for Standardization 14091 for vulnerability assessment, International Organization for Standardization and International Electrotechnical Commission 42001 for artificial intelligence management, Tanzania Data Protection Act 2022 for personal data. The conformance trail is what converts the platform from a vendor solution into national infrastructure.

1.3 What the existing platform already provides

The platform at github.com/ubuntuafyalink/afyasolar is a Next.js 16 web application written in TypeScript, backed by MySQL through Drizzle Object Relational Mapping, with NextAuth.js authentication, Cloudinary for media uploads, Twilio for WhatsApp messaging, and the GROQ Application Programming Interface for artificial intelligence features. It runs in production today for Ubuntu Facility (Kinondoni, Dar es Salaam) and other reference facilities, with a Progressive Web Application surface and several distinct dashboard routes for facility users, technicians, administrators, and an internal management panel. The platform's existing modules include facility onboarding, solar package subscription management, energy efficiency assessment, climate resilience assessment, billing and payment workflows, notifications, carbon credit calculations, and demo-token-based dashboards for showcasing to external stakeholders.

The Package Selection module currently displays eight solar package tiers from two kilowatts to thirty kilowatts, each with three pricing options (cash, installment, and Power-as-a-Service monthly fee). Each package displays its battery capacity, kilowatt capacity, and warranty terms. The Ubuntu Facility example currently shows the thirty kilowatt option selected at twenty-eight point three four million Tanzanian Shilling cash, six hundred and fifteen thousand Tanzanian Shilling per month installment, or eight hundred and fourteen thousand Tanzanian Shilling per month under Power-as-a-Service. This pricing engine is functional and already in operation.

The Energy Efficiency module currently provides a device-and-load builder with quick-add bundles for three facility archetypes (Dispensary basic, Health centre core clinical, Cold chain focus), per-device entry of category, criticality, watts, quantity, hours-per-day, and computed kilowatt-hours-per-day. Major energy use insights are computed in real time. The module captures average outage hours, facility type (on-grid or off-grid), monthly grid bill, diesel litres per day, and diesel price per litre. A Run Design and Finance Engine action triggers the sizing engine.

The Climate Resilience module currently provides a guided assessment scaffold aligned to the Climate-Resilient Primary Health Care framework, with six tabs (Hazard profile, Critical service fragility, Energy continuity and power quality, Efficiency and demand control, Readiness and response, Results and adaptation plan). The hazard profile section is the entry point and includes questions such as flood exposure with five-point scoring, optional notes, and optional evidence Uniform Resource Locator capture. The module is in Swahili and English (with the language toggle visible in the user interface). Progress is tracked and saved to the database.

1.4 The most important decisions made in this document

Decision	Rationale
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Climate data as the substrate, not telemetry	Reduces per-facility cost to zero; enables global scaling; preserves sovereignty
Three persona dashboards: Facility Manager, Operator Administrator, Impact Investor	Same data, different lenses; multiple revenue lines from one platform
WhatsApp Cloud Application Programming Interface and Short Message Service as primary channels	Matches the actual digital literacy and connectivity of facility users
Energy-as-a-Service with installation cost paid by facility, monthly fee fixed thereafter	Aligns incentives, reduces upfront barrier, predictable revenue for both parties
Artificial Intelligence as inference layer, not measurement	Generates useful predictions from sparse manual inputs plus rich climate data
Massachusetts Institute of Technology licence on the platform code	Enables third-party adoption; aligns with Digital Public Goods criteria
International standards conformance from day one	Converts platform from vendor solution to national infrastructure candidate
Localisation to Tanzania first, design for regional expansion	Maintains operational focus while preserving optionality
Eco-Pulse virtual energy benchmarking (MLR + GAM, no physical loggers)	Replaces physical audits; scales to thousands of facilities at near-zero marginal cost
Resi-Health Grid quantitative Climate Vulnerability Index	Externally verifiable from open geospatial data; citable in ministry and donor reports
Synthetic-HFE satellite-verified self-healing data integrity loop	Continuously corrects registry drift; converts the platform into a national reference
Revenue-Linked Smart-Splitter Gateway via Tigo Lipa/MCF	Daily micro-payments scale with clinic revenue; default risk effectively eliminated
Insurance Claims Receivables Escrow with NHIF assignment	Lender underwrites NHIF credit risk, not the rural clinic; unlocks lower-cost capital
Regional Cluster Insurance Protocol for decentralised O&M	Cuts isolation-tax mobilisation cost; pre-funds mid-life replacement from savings stream

Part 2 — Current State Analysis

2.1 Current technology stack

The platform at the time of this document is structured as follows.

Layer	Current Technology	Status
Web framework	Next.js 16 with App Router	Production
Language	TypeScript	Production
User Interface library	React 18	Production
Styling	Tailwind CSS 4	Production
Component primitives	Radix UI	Production
Authentication	NextAuth.js	Production
Database	MySQL 8.x via Drizzle Object Relational Mapping	Production
Media uploads	Cloudinary	Production
WhatsApp messaging	Twilio	Production
Short Message Service	SmartSMS Application Programming Interface	Production
Email	Simple Mail Transfer Protocol	Production
Artificial Intelligence	GROQ Application Programming Interface	Limited use
Service worker / Progressive Web Application	Custom implementation	Production

2.2 Current routes and their purpose

Route	User	Function
/	Anyone	Landing page; redirects authenticated users by role
/auth/signin, /auth/signup	Anyone	Authentication
/dashboard/admin	Platform Administrators	Ubuntu AfyaLink internal administration
/dashboard/facility	Facility Users	Facility-specific operations
/dashboard/technician	Field Technicians	Visit scheduling and reporting
/dashboard/management-panel	Internal Operations	Management oversight
/services/afya-solar	Anyone	AfyaSolar service marketing page
/demo/[token]	Demo Users	Token-based demo access

/demo/facility/[token]	Demo Users	Facility-specific demo dashboards
/demo/microgrid/[token]	Demo Users	Microgrid demo dashboard

2.3 Gaps identified between current platform and the target

Gap one — Climate data integration is absent

The current platform does not pull from National Aeronautics and Space Administration POWER, the European Centre for Medium-Range Weather Forecasts Re-Analysis version 5, the Climate Hazards Group Infrared Precipitation with Station data, or the Photovoltaic Geographical Information System. Every facility's solar sizing is currently computed against generic typical-meteorological-year assumptions rather than against the specific climate of the facility's coordinates. This is the single largest gap to close, because climate data is the substrate on which every subsequent intelligence capability rests.

Gap two — The three-persona dashboard architecture is incomplete

The current platform has dashboards for facility users, technicians, administrators, and a management panel. The target architecture requires three specific persona-aligned dashboards with distinct user experiences: Facility Manager (the on-site lead at the facility), Ubuntu AfyaLink Administrator (the operational lead at the operator), and Impact Investor (the funder or development finance institution viewing portfolio impact). Each persona has different access rights, different default views, different reporting formats, and different exit actions. The current management-panel route conflates several of these audiences.

Gap three — The artificial intelligence layer is rule-based, not inference-based

The platform currently uses the GROQ Application Programming Interface for limited assistance features but does not run any structured artificial intelligence models for solar yield prediction, energy demand inference, climate vulnerability classification, adaptation recommendation ranking, climate-disease early warning, or any other model family. The Climate-Resilient Primary Health Care scoring engine is rule-based on the assessment inputs. This produces correct outputs for the questions asked but does not extrapolate to predictions, forecasts, or what-if simulations.

Gap four — The international standards conformance trail is informal

The platform's documentation does not currently include a master conformance document that maps platform components to International Performance Measurement and Verification Protocol options, American Society of Heating, Refrigerating and Air-Conditioning Engineers Standard 211 audit levels, International Organization for Standardization 50001 management cycle steps, World Health Organization Guidance interventions, Intergovernmental Panel on Climate Change Sixth Assessment Report scenarios, International Organization for Standardization 14091 vulnerability methodology, or International Organization for Standardization and International Electrotechnical Commission 42001 artificial intelligence management. Without this conformance trail, the platform's outputs are credible but not citable; with it, they become citable and adoptable as national or regional reference infrastructure.

Gap five — Energy Champion programme is conceptually present but operationally absent

The platform's roadmap envisions Energy Champions as the human layer that bridges the platform to facility staff with low digital literacy. The current platform does not have a designated Energy Champion role with curriculum, certification, paid stipend infrastructure, and supervision workflows. This is the workforce layer that converts the platform from a technology product into a community-embedded service, and it is the single highest-leverage operational investment to make in the next twelve months.

Gap six — Data sovereignty posture is implicit, not explicit

The platform's data is hosted on infrastructure of Ubuntu AfyaLink's choosing and is governed by the operator's data policies. There is no explicit, published sovereignty commitment, no published Data Protection Act 2022 conformance memo, no published data classification policy, no published consent capture, no published deletion workflow. The sovereignty posture exists in practice but is not visible to prospective ministry or institutional customers who would need to see it before procurement.

2.4 What this document changes

Sections of the existing platform documented above remain. The Next.js application, the MySQL schema, the NextAuth.js authentication, the Cloudinary media handling, the WhatsApp integration via Twilio, the Package Selection module, the Energy Efficiency device-and-load builder, and the Climate-Resilient Primary Health Care assessment scaffold are all assets the re-invention preserves. The work over the next twelve months is to layer the climate data substrate on top, restructure the dashboards to the three-persona model, build the inference layer of the artificial intelligence subsystem, publish the standards conformance trail, formalise the Energy Champion programme, and publish the sovereignty commitments. None of this requires throwing away what exists. All of it requires adding the missing layers.

Part 3 — System Architecture

3.1 Architectural style

The platform is a modular monolith with extracted workers, deployed as a small set of long-running processes rather than as microservices. The single web tier serves Hypertext Transfer Protocol Secure traffic and exposes the Representational State Transfer and WebSocket Application Programming Interfaces. Computation that is asynchronous, long-running, or scheduled is extracted to worker processes that share the same code base but run different entry points. This style is chosen because it preserves operational simplicity, supports deployment to a single Virtual Private Server in the early phase (the platform's current state), and accommodates extraction to independent services in later phases without rewriting domain code.

The architectural layering, from bottom to top, is as follows.

Layer one: Ingestion. External data enters the platform here. Sources include scheduled pulls from National Aeronautics and Space Administration POWER, the European Centre for Medium-Range Weather Forecasts via Copernicus, the Climate Hazards Group Infrared Precipitation with Station data, the Photovoltaic Geographical Information System, the World Health Organization Performance Quality and Safety catalogue, the Microsoft Buildings dataset, OpenStreetMap, and the World Population dataset. Inbound webhooks from WhatsApp Cloud Application Programming Interface, Short Message Service through Africa's Talking, and mobile money providers also land here. Champion-submitted media (photographs, voice notes, documents) flow through Cloudinary.

Layer two: Data platform. Authoritative storage of all platform data. PostgreSQL version 16 with the TimescaleDB extension version 2.14 or newer for time-series climate observations and assessment histories; the pgvector extension version 0.7 or newer for vector similarity used by the Retrieval-Augmented Generation co-pilot. Redis version 7 for caching, rate-limiting, and BullMQ job queues. MinIO or Amazon Simple Storage Service for object storage of uploaded media, generated reports, and Digital Monitoring Reporting and Verification bundles. Migration path from the current MySQL schema to PostgreSQL is described in section seven of this document.

Layer three: Domain services. The eight canonical primitives — Facility, Asset, Service, Assessment, Plan, Evidence, User, Champion — and their associated business logic. Each primitive is implemented as a directory under source-library-domain with a clear public surface, internal helpers, and tests. Cross-primitive workflows live in source-library-workflows.

Layer four: Intelligence services. Stateless services that take typed input, return typed output, and may call out to model-serving endpoints. This is where the climate-data ingestion processor, the solar yield estimator, the energy demand inferer, the Climate-Resilient Primary Health Care scoring engine, the adaptation recommendation engine, the early-warning system, the generative artificial intelligence co-pilot, the solar sizing engine, the cost-effectiveness calculator, and the Digital Monitoring Reporting and Verification bundle generator live. Each intelligence service is invocable from the Application Programming Interface, from worker jobs, or from other intelligence services.

Layer five: Surfaces. Things humans and systems interact with. The Progressive Web Application served by Next.js for the three personas (Facility Manager, Ubuntu AfyaLink Administrator, Impact Investor), the WhatsApp conversational interface, the Short Message Service and Unstructured Supplementary Service Data fallback, the public data portal, and the external Application Programming Interface used by integration partners.

3.2 Process topology

The platform runs as five process types.

1. **Web server.** Runs Next.js in production mode. Serves the Progressive Web Application and the internal Application Programming Interface routes. One instance for phase one (current state); behind a load balancer with horizontal autoscaling from phase three onwards.
2. **Climate data ingestion service.** Runs scheduled pulls from National Aeronautics and Space Administration POWER, the European Centre for Medium-Range Weather Forecasts, the Climate Hazards Group Infrared Precipitation with Station data, and other free climate data sources. Stores daily climate observations and seven-day forecasts in the climate observations time-series table. Implementation in TypeScript using node-fetch and the relevant Application Programming Interface schemas, scheduled via BullMQ repeatable jobs.
3. **Background worker.** Runs BullMQ workers consuming named queues from Redis. Job types include assessment recomputation, report generation, climate data ingestion, model inference batches, model retraining, Digital Monitoring Reporting and Verification bundle creation, electronic mail and WhatsApp dispatch, mobile money reconciliation, and District Health Information System version 2 synchronisation.
4. **Scheduler.** A single instance that produces scheduled jobs (cron-style). Schedules include daily climate data ingestion at four in the morning Coordinated Universal Time, daily six-thirty local time push to Energy Champions per facility timezone, weekly Friday summary generation, weekly model inference batches, monthly Digital Monitoring Reporting and Verification bundle creation, and quarterly resilience re-assessment reminders.
5. **Machine learning serving service.** A FastAPI Python application serving the trained models. Exposes inference endpoints for solar yield, energy demand, climate vulnerability, adaptation recommendation, early warning, and the Retrieval-Augmented Generation co-pilot. Runs on a separate process (and in production, a separate node with Graphics Processing Unit) communicating with the Node.js side over Hypertext Transfer Protocol.

For phase one (currently in operation), all five process types run on a single Virtual Private Server. For phase two (anticipated at fifty to two hundred facilities), the database tier is separated. For phase three (anticipated above five hundred facilities or multi-tenant ministry deployments), each process type runs on its own node group in Kubernetes.

3.3 Technology choices, with versions

Concern	Choice	Version
Web framework	Next.js	16.x (existing)
Language	TypeScript	5.5+
Runtime	Node.js	22 Long Term Support
User Interface library	React	19
Styling	Tailwind Cascading Style Sheets	4.x
Component primitives	Radix UI	Latest

Database (target)	PostgreSQL with TimescaleDB and pgvector extensions	16.x; 2.14+; 0.7+
In-memory store	Redis	7.x
Object Relational Mapping	Drizzle Object Relational Mapping	Latest
Authentication	NextAuth (Auth.js)	5.x
Job queue	BullMQ	Latest
Object storage	MinIO or Simple Storage Service	Latest
WhatsApp messaging	WhatsApp Cloud Application Programming Interface	Native (not Twilio wrapper)
Short Message Service	Africa's Talking	Native
Embeddings	Multilingual end-to-end small five large	Latest
Local Large Language Model	Llama three point one Instruct eight billion parameters	Latest
Machine Learning operations	MLflow	2.14+
Model serving	FastAPI (Python)	Latest
Internationalisation	next-intl	Latest
Logging	pino	Latest
Observability	Prometheus, Grafana, Loki	Latest
Error tracking	Sentry	Latest
Container orchestration	Kubernetes (phase three)	1.30+

Part 4 — Database and Data Model

4.1 Migration from MySQL to PostgreSQL

The current platform runs on MySQL eight via Drizzle Object Relational Mapping. The target platform runs on PostgreSQL sixteen with the TimescaleDB extension for time-series climate observations and the pgvector extension for the retrieval-augmented generation knowledge base. The migration is performed as a one-time data import using a Python script that runs in three phases. First, the new PostgreSQL schema is created with the canonical tables documented below. Second, data is copied from MySQL to PostgreSQL with type coercion. Third, foreign key constraints and indexes are applied. The migration is idempotent and runs in approximately ten minutes against the current data volume.

4.2 Core domain tables

Facilities

The Facilities table is the central entity of the platform. Every other operational record traces back to a facility. Coordinates (latitude and longitude) are critical because they are the join key for all climate data; the platform refuses to create a facility record without valid coordinates.

Column	Type	Purpose
id	bigserial primary key	Internal identifier
uuid	uuid unique	External identifier
tenant_id	uuid foreign key	Owner tenant
external_id	text	National Health Facility Registry identifier
name	text not null	Display name
ownership_type	text	public, faith_based, private_for_profit, private_non_profit, parastatal
level	text	dispensary, health_centre, polyclinic, district_hospital, regional_hospital, specialised
region	text	Tanzanian region
district	text	Tanzanian district
ward	text	Tanzanian ward
latitude	numeric(9,6)	Decimal degrees, World Geodetic System 1984
longitude	numeric(9,6)	Decimal degrees, World Geodetic System 1984
elevation_m	integer	Metres above sea level
catchment_population	integer	Population served
roof_area_m2	integer	Roof footprint area
roof_orientation_deg	integer	Zero to 359, zero is true north

timezone	text	Africa/Dar_es_Salaam default
status	text	active, inactive, archived
metadata	jsonb	Extensible attributes

Assets

The Assets table holds every energy-consuming or energy-generating device at a facility. Solar panels, batteries, inverters, refrigerators, light fixtures, pumps, autoclaves, computers, fans — all are assets.

Column	Type	Purpose
id	bigserial primary key	Internal identifier
facility_id	bigint foreign key	Parent facility
kind	text	solar_array, inverter, battery, refrigerator, lighting_circuit, pump, autoclave, etc.
manufacturer	text	Brand
model	text	Model from nameplate
rated_power_w	numeric	Rated power in watts
capacity_kwh	numeric	For batteries
capacity_kwp	numeric	For solar arrays
installation_date	date	Installation date
condition	text	new, good, fair, poor, broken
criticality	smallint	1 to 5 (5 = mission-critical)

Climate observations (TimescaleDB hypertable)

Forty years of historical climate data plus seven-day forecasts are stored for every facility. Total volume at one thousand facilities is approximately fifteen million rows.

Column	Type	Purpose
ts	timestampz not null	Observation date
facility_id	bigint	Facility
source	text	nasa_power, era5, chirps, pvgis, ecmwf_forecast
irradiance_ghi_kwh_m2	double precision	Global horizontal irradiance
temp_max_c	double precision	Daily maximum
temp_min_c	double precision	Daily minimum
humidity_pct	double precision	Mean humidity
precipitation_mm	double precision	Daily rainfall

wind_speed_ms	double precision	Mean wind
cooling_degree_days	double precision	Derived index
wbgt_max_c	double precision	Wet-bulb globe temperature peak
forecast_horizon_days	smallint	0 for observed, 1-7 for forecast

Assessments

The Assessments table stores point-in-time facility evaluations. Energy Efficiency assessments, Climate-Resilient Primary Health Care assessments, and Workforce and Water Sanitation Hygiene assessments are all rows in this table, differentiated by the kind column.

Column	Type	Purpose
id	bigserial primary key	Internal identifier
facility_id	bigint	Parent facility
kind	text	energy_efficiency_level_1, energy_efficiency_level_2, climate_resilience, etc.
conducted_by_user_id	bigint	Auditor or Champion
conducted_at	timestampz	Assessment date
status	text	draft, submitted, verified, superseded
inputs	jsonb	Form responses
outputs	jsonb	Scores, breakdowns, recommendations

Part 5 — Integrations with External Data Sources

5.1 Integration philosophy

The platform integrates with twelve free open-source data sources. Each integration is owned by Ubuntu AfyaLink — there are no vendor cloud Application Programming Interfaces in the data flow, no proprietary protocols, no per-facility hardware investments that depend on a specific vendor. This integration list is the technical expression of the sovereignty commitment that distinguishes the platform commercially.

5.2 National Aeronautics and Space Administration POWER

What it provides: Forty years of daily climate variables (solar irradiance, air temperature, wind speed, humidity, precipitation) plus seven-day forecasts, globally, at approximately fifty kilometre spatial resolution. This is the primary climate data substrate for the platform.

Application Programming Interface acquisition: No registration required, no Application Programming Interface key required, no rate limit relevant to platform usage. Free forever, supported by the United States National Aeronautics and Space Administration as a public service.

Implementation: Scheduled BullMQ job runs daily at four in the morning Coordinated Universal Time. For each facility, the job constructs a request to power.larc.nasa.gov/api/temporal/daily/point with parameters latitude, longitude, start date, end date, and parameter list.

```
https://power.larc.nasa.gov/api/temporal/daily/point?latitude=-
6.7747&longitude=39.2664&start=20250101&end=20251231&parameters=ALLSKY_SFC_SW_DWN,T2M_MAX,T2M_
MIN,T2M,RH2M,WS2M,PRECTOTCORR&community=RE&format=JSON
```

Cost: Zero.

5.3 European Centre for Medium-Range Weather Forecasts Re-Analysis version 5 (ERA5)

What it provides: Sixty years of hourly historical reanalysis at twenty-five kilometre spatial resolution. Complement to NASA POWER with hourly granularity.

Application Programming Interface acquisition: Free registration at cds.climate.copernicus.eu. Create an account, accept the Climate Data Store terms of use, obtain a personal Application Programming Interface key.

Implementation: Python script using the cdsapi library. Scheduled monthly job pulls ERA5 data for new facilities.

Cost: Zero.

5.4 Climate Hazards Group Infrared Precipitation with Station data (CHIRPS)

What it provides: Daily precipitation data for Africa at five kilometre resolution, four decades back. Higher spatial resolution than NASA POWER for the African continent specifically.

Application Programming Interface acquisition: Free with no Application Programming Interface key required. Data downloadable from chc.ucsb.edu/data/chirps.

Cost: Zero.

5.5 Photovoltaic Geographical Information System (PVGIS)

What it provides: Site-specific solar resource modelling including horizon shading from terrain. European Commission Joint Research Centre's photovoltaic resource estimator.

Application Programming Interface acquisition: Free, no Application Programming Interface key. Available at re.jrc.ec.europa.eu/pvg_tools/en/.

Cost: Zero.

5.6 pvlib-python

What it provides: Reference open-source solar simulation library, maintained by Sandia National Laboratories. Implements every published solar resource model. The substrate of every credible solar simulation.

Application Programming Interface acquisition: Apache two licence, free, install via pip install pvlib.

Cost: Zero.

5.7 WorldPop

What it provides: Open population density data at one hundred metre resolution globally. Lets the platform compute the catchment population of any facility from coordinates alone.

Application Programming Interface acquisition: Free, no Application Programming Interface key. Data available at worldpop.org.

Cost: Zero.

5.8 Microsoft Buildings dataset

What it provides: Open dataset of building footprints across Africa derived from satellite imagery. For Tanzania approximately twenty million building footprints with polygons and approximate areas.

Application Programming Interface acquisition: Free, no Application Programming Interface key. Open Data Commons Open Database Licence. Download from github.com/microsoft/GlobalMLBuildingFootprints.

Cost: Zero.

5.9 OpenStreetMap

What it provides: Road network, electrification proxies, water bodies, administrative boundaries, points of interest.

Application Programming Interface acquisition: Free under the Open Database Licence. Available via Overpass Application Programming Interface or via bulk Geographic Information System Document File downloads.

Cost: Zero.

5.10 World Health Organization Performance Quality and Safety catalogue

What it provides: Reference database of prequalified vaccine cold chain equipment with specifications.

Application Programming Interface acquisition: Free, public, available at extranet.who.int/pqweb/devices. Updated quarterly by the World Health Organization.

Cost: Zero.

5.11 Intergovernmental Panel on Climate Change Sixth Assessment Report emission factors

What it provides: Authoritative carbon dioxide equivalent factors for diesel combustion, petrol combustion, kerosene combustion, charcoal, natural gas, and grid electricity.

Implementation: Static reference data in the platform code base. Versioned.

Key factors: Diesel two point sixty-eight kilograms carbon dioxide equivalent per litre, petrol two point thirty-one kilograms carbon dioxide equivalent per litre, grid electricity for Tanzania zero point five five kilograms carbon dioxide equivalent per kilowatt-hour.

Cost: Zero.

5.12 Sentinel-2 satellite imagery

What it provides: Ten metre resolution multispectral imagery globally, refreshed every five days. Useful for verifying installed solar arrays from satellite.

Application Programming Interface acquisition: Free with a Copernicus Open Access Hub account at scihub.copernicus.eu.

Cost: Zero.

5.13 WhatsApp Cloud Application Programming Interface

What it provides: Direct Application Programming Interface access to Meta's WhatsApp messaging infrastructure. Replaces the current Twilio wrapper to reduce messaging cost and increase functionality.

Application Programming Interface acquisition: Free Application Programming Interface, with per-message charges. Register at developers.facebook.com.

Cost: Approximately zero point zero zero five United States dollars per service message.

5.14 Africa's Talking Short Message Service and Unstructured Supplementary Service Data

What it provides: Short Message Service delivery to Tanzanian mobile numbers and Unstructured Supplementary Service Data menu hosting.

Application Programming Interface acquisition: Register at africastalking.com, obtain Application Programming Interface key, configure shortcode.

Cost: Short Message Service approximately twenty Tanzanian Shilling per message; Unstructured Supplementary Service Data session approximately ten Tanzanian Shilling per session.

5.15 Summary: free integrations list

Integration	Free?	API Key Required?
NASA POWER	Yes	No
ECMWF ERA5	Yes	Free registration
CHIRPS	Yes	No
PVGIS	Yes	No
pvlb-python	Yes	Library
WorldPop	Yes	No
Microsoft Buildings	Yes	No
OpenStreetMap	Yes	No
WHO PQS	Yes	No
IPCC emission factors	Yes	Static data
Sentinel-2	Yes	Free registration
GADM	Yes	No
WhatsApp Cloud	Per-message	Yes
Africa's Talking	Per-message	Yes
ClickPesa / Selcom	Per-transaction	Yes

5.16 Open-Meteo (Forecast and Historical Weather)

What it provides: Free, no-key, hourly forecast and historical weather data for any latitude and longitude globally. Used by the Eco-Pulse Engine for the localised ambient-temperature feed that drives the cooling-load adjustment in the Energy Performance Index calculation.

Application Programming Interface acquisition: Free, no Application Programming Interface key, no registration. Endpoint: api.open-meteo.com.

Cost: Zero.

5.17 Visible Infrared Imaging Radiometer Suite Day-Night Band (VIIRS)

What it provides: Monthly night-time lights radiance imagery at approximately five-hundred metre resolution globally, refreshed monthly. The primary data feed for the Synthetix-HFE module. Each facility coordinate yields a monthly radiance value that is tracked as a time-series; significant deviations from a facility's historical baseline are flagged as candidate data-integrity anomalies.

Application Programming Interface acquisition: Free, through the United States Earth Observation Group at the Colorado School of Mines, distributed through the National Oceanic and Atmospheric Administration. Monthly composite tiles downloadable from eogdata.mines.edu/products/vnl.

Cost: Zero.

5.18 Tigo Pesa Lipa Merchant Application Programming Interface

What it provides: Direct integration with the Tigo Pesa merchant payment service. Powers the Revenue-Linked Smart-Splitter Gateway by issuing a dedicated Lipa merchant Till number per facility, applying an automated split rule at the moment of payment receipt before settlement to the clinic, and producing real-time reconciliation events the platform consumes through webhooks.

Application Programming Interface acquisition: Commercial partnership with Tigo Tanzania. Onboarding requires registration as a corporate merchant, due diligence, and signature of the Tigo Pesa Lipa merchant agreement. Once provisioned, the platform receives a sandbox key for development and a production key for live transactions. Webhooks fire on every transaction.

Cost: Per-transaction. Tigo Pesa charges a tiered merchant discount rate on the gross payment; the Smart-Splitter applies the contractual split coefficient on top of the net settlement. Exact rates depend on the volume tier the platform negotiates with Tigo Tanzania.

5.19 National Health Insurance Fund Claims Application Programming Interface

What it provides: Programmatic integration with the National Health Insurance Fund claims and disbursement system. Used by the Insurance Claims Receivables Escrow to file the Notice of Assignment of Receivables, to monitor incoming claim payouts, and to reconcile the escrow account against the underlying claims.

Application Programming Interface acquisition: The National Health Insurance Fund operates a partner integration programme for accredited providers. Onboarding requires a signed Memorandum of Understanding, a regulated trust account at a Tanzanian commercial bank, and validation of the facility's status as a National Health Insurance Fund-empanelled provider.

Cost: No per-transaction fee from the National Health Insurance Fund. Standard bank fees apply on the trust account that receives the assigned receivables.

Part 6 — International Standards (Localised to Tanzania)

6.1 Energy efficiency standards

ASHRAE Standard 211-2018

The global reference standard for commercial building energy audits. Defines three audit levels. Level one is walk-through, low-cost, identifies low-cost opportunities. Level two is energy survey and analysis, the standard professional audit. Level three is detailed analysis of capital-intensive modifications, investment-grade. The platform adopts this taxonomy directly. Audit reports cite Standard 211 explicitly on the first page. Localised to Tanzania, this means the audit level is named to match the Tanzania Ministry of Energy guidance on facility energy assessments, while preserving the underlying methodology integrity.

ISO 50001:2018 Energy Management Systems

The process standard for systematic energy management. Defines the Plan-Do-Check-Act cycle as it applies to energy. Conformance to ISO 50001 is what allows the platform to be marketed as an Energy Management System, not merely an audit tool. The certification pathway is optional and expensive; conformance is free and signals professional discipline. The platform exposes an ISO 50001 Workspace per facility for larger-facility tier customers (faith-based hospitals, district hospitals).

ISO 50002:2014 Energy Audits

The audit-specific companion to ISO 50001. Defines audit planning, data collection, analysis, and reporting at a higher level of generality than Standard 211. Adopted as the procedural backbone.

ISO 50006:2014 Energy Baselines and Energy Performance Indicators

The technical standard for defining baselines and indicators. Critical because every savings claim depends on the integrity of the baseline. Specifies how to normalise the baseline for relevant variables (climate, occupancy, services delivered) so that comparisons are like-for-like. The platform's counterfactual baseline model encodes this normalisation logic.

International Performance Measurement and Verification Protocol Core Concepts 2022

Defines the four options (A, B, C, D) for measuring and verifying energy savings. The platform tags every Energy Conservation Measure with its applicable IPMVP Option, executes verification according to the option's requirements, and labels every savings claim with the methodology used. International Performance Measurement and Verification Protocol is what turns the Digital Monitoring Reporting and Verification bundle from self-attestation into a credible third-party-verifiable instrument.

WHO Performance Quality and Safety E03 — Refrigerators

The vaccine refrigerator specification standard. Every cold-chain refrigerator the platform recommends must be on the WHO PQS E03 prequalified list. This is non-negotiable for any facility that runs the Expanded Programme on Immunization cold chain.

6.2 Climate resilience and adaptation standards

WHO Operational Framework for Building Climate Resilient Health Systems (2015)

The foundational document. Defines ten components of a climate-resilient health system: leadership and governance; health workforce; vulnerability, capacity and adaptation assessment; integrated risk monitoring and early warning; health and climate research; climate-resilient and sustainable technologies and infrastructure; management of environmental determinants; climate-informed health programmes; emergency preparedness and management; climate and health financing. The platform addresses approximately four of them directly.

WHO Guidance for Climate Resilient and Environmentally Sustainable Health Care Facilities (2020)

The facility-level document. Defines four fundamental requirements (health workforce; water sanitation hygiene and healthcare waste; sustainable energy services; infrastructure technologies and products) and enumerates twenty-four specific interventions across these four pillars. The Climate-Resilient Primary Health Care framework in the platform maps dimension by dimension to this guidance.

Intergovernmental Panel on Climate Change Sixth Assessment Report Working Group II Africa Chapter

The climate science reference for African adaptation. Contains the projected climate scenarios for Tanzania and the broader region, the vulnerability assessments, and the adaptation pathway recommendations. The platform's climate projections cite the IPCC Sixth Assessment Report Shared Socioeconomic Pathway 2-4.5 and Shared Socioeconomic Pathway 5-8.5 scenarios specifically.

ISO 14091:2021 Climate Change Adaptation Vulnerability and Risk Assessment

The formal standard for how to conduct a climate change vulnerability and risk assessment. Defines the methodology: hazard identification, exposure assessment, sensitivity assessment, adaptive capacity assessment, risk evaluation, adaptation option appraisal. The platform's Climate-Resilient Primary Health Care framework dimensions map onto these stages.

Tanzania National Adaptation Plan and Tanzania Health Sector Strategic Plan version 5

Tanzania submitted its National Adaptation Plan to the United Nations Framework Convention on Climate Change secretariat with health as one of the priority sectors. The Tanzania Health Sector Strategic Plan version 5 is the operational planning document. The platform explicitly aligns its facility-level interventions with the national-level adaptation priorities and the strategic plan indicators.

6.3 Data and interoperability standards

HL7 Fast Healthcare Interoperability Resources Release 5

The international healthcare interoperability standard. The platform exposes Facility, Service, Observation, Device, and Organization resources via FHIR endpoints. Implementation is via a HAPI Fast Healthcare Interoperability Resources sidecar.

District Health Information System version 2 Web Application Programming Interface

Tanzania's Health Management Information System runs on DHIS2. The platform's facility-level data flows into DHIS2 in both directions: pulling the canonical facility identifiers from the National Health Facility Registry and pushing the platform's Climate-Resilient Primary Health Care indicators as DHIS2 data elements.

Tanzania Data Protection Act 2022

The local data protection law. Personal data of facility staff, patients, and Energy Champions is handled according to the Act's requirements: explicit consent, purpose limitation, data minimisation, security, breach notification. Implementation is via the data classification system in the platform's data layer plus consent capture in user onboarding plus application-level encryption for personally-identifying-information fields.

6.4 Artificial Intelligence standards

ISO/IEC 42001:2023 Artificial Intelligence Management System

Published in late 2023, the first formal management-system standard for Artificial Intelligence. Defines the discipline for managing Artificial Intelligence systems: governance, risk management, training data management, model lifecycle, monitoring, incident response. The platform adopts conformance to ISO 42001 as the Artificial Intelligence governance baseline.

United States National Institute of Standards and Technology Artificial Intelligence Risk Management Framework 1.0

The most thoroughly worked-out Artificial Intelligence risk-management framework currently available. Defines four core functions: Govern, Map, Measure, Manage. The platform's Artificial Intelligence governance documentation is structured around this framework.

WHO Ethics and Governance of Artificial Intelligence for Health (2021)

The World Health Organization's specific guidance for Artificial Intelligence in health applications. Defines six principles: protect autonomy; promote human wellbeing and safety; ensure transparency; foster responsibility and accountability; ensure inclusiveness and equity; promote responsive and sustainable Artificial Intelligence. The platform's Artificial Intelligence governance document is structured around these six principles.

Model Cards and Datasheets for Datasets

Every model the platform deploys ships with a Model Card stating intended use, training data, performance metrics stratified by relevant subgroups, known failure modes, ethical considerations, and contact information. Every training dataset ships with a Datasheet stating motivation for collection, composition, collection process, preprocessing, labelling process, and use restrictions.

Part 7 — Data Collection Specifications

7.1 The fifteen-parameter Minimum Viable Audit

The platform's energy audit is captured through fifteen parameters that an Energy Champion can collect via WhatsApp in approximately three hours of cumulative time spread across two weeks. These fifteen parameters carry ninety-five percent of the diagnostic value of a full audit and are the platform's structural answer to scaling assessment without scaling field-officer headcount.

Tier 1 — Always captured (Champion via WhatsApp, ten minutes)

Parameter	Capture Mode	Validation
Facility name and GPS coordinates	Auto from device GPS	Within Tanzania bounding box
Facility type	Multiple choice	From controlled vocabulary
Number of rooms	Single integer	Between 1 and 50
Number of staff	Single integer	Between 1 and 500
Patients per month	Single integer	Between 100 and 100000
Services offered	Multi-select checklist	From WHO service taxonomy
Current energy sources	Multi-select	grid, diesel, solar, none
Monthly grid bill	Three photographs	Cloudinary OCR extracts kWh, TSh
Monthly diesel cost	Single number plus photograph of receipt	TSh per month
Number of refrigerators with ages	Two integers (count, average age)	Reasonable range
Outages per month	Single integer	Self-reported
Vaccine spoilage past 12 months	Single TSh estimate	Optional
After-hours load observation	Yes/No plus voice note	Champion observation
Generator runtime ratio	Computed	Generator hours divided by outage hours
Climate exposure	Auto-computed	From NASA POWER for facility coordinates

7.2 The three-output audit report

The fifteen parameters produce three specific outputs. Not seven outputs, not a forty-page report; three outputs. Each is anchored to a specific decision the facility manager, district health officer, or funder needs to make.

Output one — Waste eliminated

Headline: total monthly energy spend, broken down by source (grid, diesel, kerosene, candles, batteries). Within that total, the portion identified as waste that can be eliminated without changing service delivery: specifically the cost of inefficient refrigerators, the cost of after-hours phantom loads, the cost of generator over-runtime, the cost of inflated grid bills (if any), and the cost of vaccine spoilage.

Output two — Monthly cash saved

Headline: under the recommended Energy-as-a-Service contract, the monthly fee Ubuntu AfyaLink would charge versus the facility's current monthly spend. The cash saving from month one. Cumulative saving over the contract term (typically seven years). The asset-transfer event at end of term. This is the financial case in language the facility manager understands.

Output three — Cost per service-hour reduced

Headline: the cost per hour of facility operation with cold chain functional, lighting available, and critical equipment powered. Today versus after intervention. This is the platform's signature metric and it is the one that maps cleanly to health outcome language used by ministries and funders.

Part 8 — Three Persona Dashboards

8.1 Design philosophy: same data, three lenses

The platform serves three primary user personas. The Facility Manager (the nurse-in-charge or designated Energy Champion at the facility) uses the platform daily for operational decisions about her facility. The Ubuntu AfyaLink Administrator (the operational lead at Ubuntu AfyaLink) uses the platform daily for portfolio management across many facilities. The Impact Investor (the funder, Development Finance Institution analyst, or government planner) uses the platform periodically for outcome verification and capital allocation. The platform serves all three from one data substrate, but each persona sees a fundamentally different interface.

The architectural principle is that the data is canonical and shared, but the interface is persona-specific. The Facility Manager sees her facility in detail; the Administrator sees all facilities in aggregate with drill-down; the Impact Investor sees impact metrics with verification links. Same data underneath, different lenses on top. This is the structural pattern that makes the platform commercially valuable to multiple audiences from one engineering investment.

8.2 Facility Manager dashboard

Persona profile

The Facility Manager is a nurse-in-charge or designated Energy Champion at a single facility. She has working-level Swahili literacy and functional but uncomfortable English. She operates on a Tecno Spark 8C or equivalent budget smartphone with two gigabytes of random access memory. She has between five hundred megabytes and two gigabytes of mobile data per month. She has between two and six minutes per day available for the platform. Her phone is shared with family members in the evenings. The internet at her facility works between eight in the morning and noon, and four to seven in the afternoon, on most days but not all. WhatsApp works through everything because it is on her data plan; Short Message Service reaches her when WhatsApp does not.

Home page — the Today surface

The Facility Manager's home page answers three questions in five seconds with no scrolling on a 5.5-inch screen: are the vaccines safe right now, how much power will I have today, and is there something I need to do. The page is structured as three large cards stacked vertically.

Card one displays the fridge status. The headline is a single Swahili word — SALAMA (safe), HATARI (danger) — in 64-point display type. Below the word, the current temperature in degrees Celsius in 18-point type. Below that, the last-checked timestamp in muted small type. The card uses the success colour palette (green) when safe and the danger palette (red) when unsafe. When the fridge is over-temperature, the entire page reorganises with this card taking the full hero position and a single Niambie tatizo (Tell me the problem) call-to-action appearing for one-tap troubleshooting flow.

Card two displays the power forecast for today. Three lines: expected hours of power based on the seven-day climate forecast and the facility's specific solar resource profile; current battery State of Charge as a percentage; expected solar today (sunny, cloudy, partly). The expected hours estimate is derived from the climate forecast plus the facility's pvlib-modelled generation profile.

Card three displays the pending tasks for today. Most days this card is empty. When something requires action — a technician visit scheduled, a meter reading due, a quarterly assessment due, an alert that requires Champion response — the task appears here with a single tap to start the relevant workflow.

Bottom navigation

The bottom navigation has exactly five items: Leo (Today, home), Friji (Fridge, detail page), Umeme (Power, detail page), Ripoti (Reports, submit daily reports), and a help-and-call button. Five tabs because the human eye scans five in one fixation; six would not. Tap targets are minimum forty-four by forty-four pixels per Web Content Accessibility Guidelines version 2.2 Level AA.

Page: Friji detail

A 24-hour line chart of fridge interior temperature, with a green band between two and eight degrees Celsius. Built with Recharts (already a dependency), server-rendered to Scalable Vector Graphics on first paint, hydrated to interactive on the client. Below the chart, an events list: door openings, temperature excursions, manual readings, maintenance visits. A Pima joto sasa (measure temperature now) button opens the camera to scan the fridge logger display via optical character recognition and saves a manual reading. A Tatizo na friji (problem with the fridge) button opens a guided troubleshooting flow.

Page: Umeme detail

Current power source displayed as a single large icon with the active source coloured. Current power flow displayed as a Sankey-style diagram (solar production, household consumption, battery State of Charge, grid status). A 24-hour stacked area chart of power by source. A forecast for the next twelve hours showing expected power source and expected battery State of Charge over time. A service-hours-remaining estimate: At current usage, your facility can deliver critical services until 06:47 tomorrow morning.

Page: Ripoti (Daily Reports)

The most common task. The form has exactly four required fields, each large enough to fit on the screen on its own. Wagonjwa walioonana leo (Patients seen today): single integer input with numeric keypad. Watoto walipata chanjo (Children vaccinated today): single integer input. Mama waliyojifungua (Mothers who delivered today): single integer input. Mafuriko au tatizo lolote (Floods or any problem): voice note recorder. Each step uses the entire screen. The user can tap-to-advance or swipe-to-advance. Voice input enabled on every numeric field. Submission triggers local IndexedDB write (instant), Application Programming Interface call (background), and on success the next-day task scheduling plus DHIS2 sync queueing.

Page: WhatsApp interaction surface

The Facility Manager's WhatsApp surface is the daily-touch surface. At six-thirty in the morning local time, she receives a Swahili message addressing her by preferred name and reporting fridge status, expected power today, and any pending tasks. She can reply with voice notes, photographs, or text. The platform's WhatsApp number is the same across all facilities; the per-facility context is established by the registered phone number. Voice notes are transcribed by Whisper-large-v3 with Swahili language hint; the transcription drives ticket routing and co-pilot retrieval.

8.3 Ubuntu AfyaLink Administrator dashboard

Persona profile

The Ubuntu AfyaLink Administrator is the operational lead at Ubuntu AfyaLink — Anodi as Chief Executive Officer, Idda as Chief Financial Officer, Albert as Chief Technology Officer, or one of the future operations and partnerships staff. She is highly literate in English and Swahili, operates on a laptop with reliable connectivity, and uses the platform for several hours per day. She manages the entire portfolio of facilities, monitors operational metrics, approves interventions, manages technician scheduling, tracks revenue and costs, prepares funder reports, and onboards new facilities.

Home page — Portfolio overview

The home page is a single dense view with five panels arranged in a grid.

Panel one (top-left, large) displays the portfolio map. Every facility is a marker on a map of Tanzania, coloured by current operational status (green for healthy, amber for attention needed, red for critical). Tapping a marker opens a side sheet with that facility's summary. The map uses MapLibre with OpenStreetMap tiles. Tile caching is aggressive; the Administrator can pan and zoom offline.

Panel two (top-right) displays the operational metrics headline: facilities active, total energy delivered this month, outages prevented this month, carbon dioxide equivalent avoided this month. Each with a sparkline trend across the past twelve months.

Panel three (middle-left) displays the ranked attention list: the five facilities most needing administrator attention today, with the reason (predictive cold-chain failure expected within 72 hours, payment overdue more than 30 days, technician visit overdue, Champion has not reported in 14 days, climate alert active for region).

Panel four (middle-right) displays the financial summary: monthly recurring revenue from Energy-as-a-Service contracts, accounts receivable, monthly operating costs, gross margin, all as small numbers with trend arrows.

Panel five (bottom, wide) displays the recent activity log: new facilities onboarded this week, completed assessments, technician visits, payments received, evidence bundles published. Configurable filters per Administrator preference.

Facilities page

A filterable, sortable table of all facilities in the portfolio. Columns: facility name, region, district, type, current Climate-Resilient Primary Health Care score, current energy status, contract status, last assessment date, last payment date, next action. Filters: by region, by type, by score range, by status, by intervention category. Export to comma-separated values for ad-hoc analysis.

Technicians and field operations page

A calendar view of technician visits across the portfolio, with the ability to schedule new visits, reassign visits, and view technician utilisation. Each scheduled visit is colour-coded by type (installation, maintenance, audit, emergency response). Visit reports filed by technicians are linked to the calendar entries.

Assessments and intervention queue

All assessments across the portfolio in submitted-status awaiting review. The Administrator can approve them, request revisions, or escalate to the Chief Executive for review. Approved assessments trigger the financial-modelling step and become the basis for intervention proposals to the facilities.

Financial operations page

Monthly invoice generation, payment reconciliation against mobile money provider records, accounts receivable management, expense tracking, and a profit-and-loss view aggregated across the portfolio. Integration with the mobile money Application Programming Interface for automatic Customer-to-Business payment prompts and reconciliation.

Reporting page

Pre-built report templates for the funders Ubuntu AfyaLink has relationships with: Medical Credit Fund quarterly impact report, Christian Social Services Commission portfolio report, Sustainable Energy for All Powering Healthcare reporting, ASCENT Programme reporting. Each template populates from the platform's data and is exportable as a Portable Document Format. New report templates can be added without engineering involvement once the template editor is built.

8.4 Impact Investor dashboard

Persona profile

The Impact Investor is the funder, Development Finance Institution analyst, ministry planner, or foundation programme officer who has invested or is considering investing in Ubuntu AfyaLink, its portfolio, or its sector. He or she is highly literate in English, operates on a corporate-issue laptop with full connectivity, uses the platform infrequently (monthly to quarterly), and needs verifiable, audit-grade outcome data when accessing it. The Impact Investor view is the platform's commercial credibility surface; this is where due diligence happens.

Home page — Impact summary

A single dense page with four sections.

Section one is the hero metrics. Four large cards: facilities active in portfolio, total catchment population served, vaccines protected this year, carbon dioxide equivalent avoided this year. Each with a trend chart across the period of the funder's involvement. Each with a single sentence of context underneath citing the methodology and the verification source.

Section two is the cost-per-outcome breakdown. Dollar per child served, dollar per service-hour gained, dollar per tonne of carbon dioxide equivalent avoided, dollar per percentage-point of Climate-Resilient Primary Health Care score improved. Each computed from the platform's data with methodology footnotes.

Section three is the verification surface. A list of Digital Monitoring Reporting and Verification bundles published in the period, each downloadable as JavaScript Object Notation Linked Data or as a rendered Portable Document Format. Each bundle is cryptographically signed by Ubuntu AfyaLink and optionally counter-signed by the facility. The Impact Investor can verify any bundle independently through the platform's verification endpoint.

Section four is the geographic distribution. A map of the portfolio with each facility coloured by current Climate-Resilient Primary Health Care score, with the ability to filter by region, by type, by intervention category.

Drill-down: per-facility impact

From the home page, the Impact Investor can drill into any single facility to see its specific impact contribution: the catchment population, the baseline situation pre-intervention, the interventions deployed, the outcomes observed, the avoided emissions, the financial flows, the Champion roster, and the climate exposure profile. Every metric on the page links to its supporting data and its methodology citation. The Impact Investor leaves the page understanding both what changed and how the platform measured it.

Comparison view

The Impact Investor can compare this portfolio's metrics against anonymised peer benchmarks (when other operators are on the platform and have opted into benchmark sharing). This view shows the operator's portfolio metrics versus the peer median across cost per service-hour, cold-chain uptime, score improvement rate, and intervention pay-back period. This is the surface that supports the Impact Investor's allocation decisions across multiple operators.

Export and reporting

The Impact Investor can export the data underlying any view as comma-separated values for the Impact Investor's own analysis and reporting. The Impact Investor can also generate pre-formatted Portable Document Format reports aligned to the Impact Investor's own reporting cycle (quarterly, annually) that can be incorporated directly into the Impact Investor's internal documents.

Part 9 — Energy Efficiency Module (Re-Invented)

9.1 Current state

The existing platform's Energy Efficiency module has a working device-and-load builder with quick-add bundles for three facility archetypes (Dispensary basic, Health centre core clinical, Cold chain focus), per-device entry of category, criticality, watts, quantity, hours-per-day, and computed kilowatt-hours-per-day, plus a Major Energy Use insights panel and a Run Design and Finance Engine action. The module is functional and is the primary site of facility energy data entry today.

9.2 The fifteen-parameter Minimum Viable Audit as the entry point

The re-invented module enters through a guided WhatsApp-first audit flow that captures the fifteen parameters specified in Part Seven. The Energy Champion is led through the audit in two-minute increments spread across approximately two weeks. The flow is staged as follows.

Week one, day one. The Champion receives a welcome WhatsApp message introducing herself to the platform and asking permission to begin the audit. She accepts. The platform sends the first question: Tafadhali piga picha ya kila chumba (please photograph each room). She walks through the facility and sends seven to ten photos. The vision model processes the photos and produces a draft equipment inventory.

Week one, day two. The Champion receives a follow-up message: We identified twelve devices from your photos. Please confirm or correct. She reviews the draft list and corrects any misidentifications. The inventory is locked.

Week one, day three. Tafadhali tuma picha ya bili tatu za hivi karibuni za TANESCO (please send photographs of three recent grid bills). Cloudinary optical character recognition extracts the kilowatt-hours, the demand charges, the time-of-use breakdown if present.

Week one, day four. Tafadhali tuma ankara za mafuta za miezi mitatu iliyopita (please send fuel receipts for the past three months). Extraction of monthly diesel cost.

Week one, day five through fourteen. Daily outage and waste logging. Each evening: Leo umeme ulikatika? Saa ngapi? (today did the power go out? at what time?). The Champion sends short voice notes or text replies. The platform builds the outage log over the two-week period.

End of week two. The platform generates the three-output audit report — waste eliminated, monthly cash saved, cost-per-service-hour reduced — and sends it as both a WhatsApp message summary and a downloadable Portable Document Format. The Champion shares the report with the Nurse-in-Charge for review. If the facility wants to proceed, the conversation moves to the financial modelling and Energy-as-a-Service contract proposal.

9.3 Synthetic load profile generation

With the equipment inventory complete, the platform constructs a synthetic facility load profile rather than relying on physical sub-metering. The synthetic profile uses the following methodology, executed in the Python sub-project.

Per device, the platform computes initial estimated daily consumption as rated power times hours of operation per day, plus a standby allowance (five percent for refrigerators and autoclaves, ten percent for computers and laboratory analysers, zero for lighting and pumps). For refrigerators and air conditioners, an ambient-temperature multiplier is applied from the climate observations table for the facility's location: one point zero at twenty-five degrees Celsius, approximately one point five at thirty degrees, approximately two point two at thirty-five degrees.

Per device, the operational schedule template (continuous-twenty-four-hours, office-hours-eight-to-five, scheduled-evening-only, scheduled-during-procedures, intermittent-as-needed, etc.) determines when during a typical day the device is active. Aggregated across devices, the platform produces an hour-by-hour load profile for the facility on a typical day in each season.

The synthetic profile is calibrated against the three months of utility bill data captured during the audit. If the synthetic profile predicts five hundred kilowatt-hours per month and the bill says six hundred and twenty, the platform scales the profile by approximately twenty-four percent and flags the gap as an unaccounted load for Champion follow-up. With three or four iterations of calibration, the load profile matches measured monthly consumption within five to ten percent.

9.4 ASHRAE Standard 211 Level 1, 2, 3 differentiation

The platform's audit module supports all three levels of the American Society of Heating, Refrigerating and Air-Conditioning Engineers Standard 211.

Level one is the default: the fifteen-parameter Minimum Viable Audit. Walk-through, photograph-based, three hours of Champion time, free to the facility. Identifies low-cost interventions.

Level two is a chargeable upgrade for facilities that want detailed engineering work: sub-metering on critical circuits for fourteen days (in cases where the facility is large enough to justify the cost of installing temporary loggers), twelve-month bill analysis, formal Standard 211 Level 2 report, investment-grade sizing scenarios. Priced approximately four hundred thousand to one million two hundred thousand Tanzanian Shilling per facility.

Level three is a chargeable engagement for Development Finance Institution lending preparation: full detailed analysis with calibrated energy model in pvlib and custom thermal model, multi-scenario analysis under four climate projections, sensitivity analysis on key cost drivers, six-to-eight-week engagement producing an investment-grade report. Priced one million eight hundred thousand to four million Tanzanian Shilling per facility, typically paid by the Development Finance Institution rather than the facility.

9.5 The Energy Conservation Measures catalogue

Energy Conservation Measures (Energy Conservation Measures) are the interventions the platform recommends. The catalogue is a versioned YAML file at docs/standards/ecms.yaml containing approximately fifty Energy Conservation Measures relevant to Tanzanian primary healthcare facilities. Each entry contains: code, title in English and Swahili, category, description, prerequisites, cost formula (typically a function of facility size), savings formula, applicable International Performance Measurement and Verification Protocol option, expected lifetime, expected resilience gain by Climate-Resilient Primary Health Care dimension, World Health Organization intervention reference, supporting evidence citations.

Example Energy Conservation Measures in the catalogue include: light-emitting diode lighting retrofit (Energy Conservation Measure 001, applicable to almost every facility), solar direct drive vaccine refrigerator replacement (Energy Conservation Measure 002), critical-load circuit isolation (Energy Conservation Measure 003), cool-roof coating (Energy Conservation Measure 004), variable-frequency drive on water pump (Energy Conservation Measure 005), power factor correction at main panel (Energy Conservation Measure 006), surge protection and voltage stabilisation (Energy Conservation Measure 007), and so on through the catalogue.

9.6 The Eco-Pulse Virtual Energy Performance Index Engine

The Eco-Pulse Engine is the platform's lightweight, software-only energy efficiency benchmarking layer. It exists because conducting physical energy efficiency audits on hundreds of rural clinics is economically unworkable: travel and engineering fees alone consume thousands of United States dollars per assessment, and the resulting Portable Document Format reports become stale within months. Eco-Pulse replaces physical auditors with statistical inference grounded in proxy variables and live climate data.

The engine accepts a minimum data set entered through a mobile form by the district health officer or Energy Champion: facility tier (dispensary, health centre, hospital), gross floor area in square metres, bed count, monthly outpatient volume, and an appliance inventory drawn from controlled vocabulary. The form takes approximately seven minutes to complete on a Tecno Spark-class smartphone. No physical loggers, no sub-meters, no engineering site visits are required to produce the first benchmark. These inputs are paired with the facility's localised ambient-temperature feed from Open-Meteo and the historical climate profile from National Aeronautics and Space Administration POWER. The engine then runs two statistical models in tandem. A Multiple Linear Regression model produces a first-pass expected baseline consumption from the facility's characteristics. A Generalised Additive Model captures the non-linear interaction between ambient temperature and cooling-load demand, which a linear model cannot represent because cooling load grows non-linearly above approximately twenty-eight degrees Celsius. The two model outputs are blended through stacking to produce a single Energy Performance Index per facility.

The Energy Performance Index is expressed as the ratio of observed to expected monthly consumption, adjusted for the facility's climate. A value of one means the facility consumes the energy expected for its tier, services, and climate. A value above one point three flags significant inefficiency relative to peer facilities; below zero point eight may indicate under-reporting or measurement error. With each flag the engine surfaces specific diagnostic hypotheses: "Clinic X consumes thirty-five percent more overnight energy than its peer baseline; the most probable cause is cooling-insulation degradation on the vaccine refrigerator or an unaccounted after-hours load."

Implementation lives in the Python sub-project at source-machine-learning-models-eco-pulse. The model is retrained quarterly as new facilities and assessment cycles accumulate. The Energy Performance Index is exposed through the Application Programming Interface and rendered on the Ubuntu AfyaLink Administrator dashboard as a sortable column on the Facilities page, with anomaly flags that link directly to suggested intervention sets from the Energy Conservation Measures catalogue.

Part 10 — Climate Resilience Module (CRiPHC v2.0)

10.1 Current state

The existing platform's Climate Resilience module is the most architecturally complete climate-health scoring tool in the African primary healthcare space at the time of this writing. The six-tab structure (Hazard profile, Critical service fragility, Energy continuity and power quality, Efficiency and demand control, Readiness and response, Results and adaptation plan) maps cleanly to the World Health Organization Guidance four pillars and to the International Organization for Standardization 14091 vulnerability and risk assessment methodology. The five-point scoring per question, the evidence-link capture, the Swahili language toggle, the progress tracking, and the database persistence are all working. This is a genuine asset to preserve and extend.

10.2 The Climate-Resilient Primary Health Care framework version 2.0

The re-invented framework expands the existing five dimensions (Hazard Exposure Score, Critical Service Fragility, Energy Continuity and Power Quality, Efficiency and Demand Control, Readiness and Response Capacity) with two new dimensions to fully cover the World Health Organization Guidance four pillars: Workforce (the staff's training, retention, and protection during climate stress) and Water Sanitation Hygiene and Waste (the facility's water security, sanitation infrastructure, and healthcare waste management). The composite Resilience Capacity Score is the weighted average of these seven dimensions.

Dimension	Code	Default Weight
Hazard Exposure Score	HES	0.15
Critical Service Fragility	CSF	0.20
Energy Continuity and Power Quality	ECPQ	0.15
Efficiency and Demand Control	EDC	0.10
Readiness and Response Capacity	RRC	0.10
Workforce	W	0.15
Water Sanitation Hygiene and Waste	WW	0.15

10.3 Climate hazard data integration

The single largest enhancement to the Climate Resilience module is the integration of National Aeronautics and Space Administration POWER and European Centre for Medium-Range Weather Forecasts Re-Analysis version 5 climate data. The Hazard Exposure Score, previously dependent on Champion self-reporting (Flood exposure around the facility: Low / Medium / High), now combines Champion self-reporting with the forty-year climate exposure profile for the facility's coordinates. The platform produces a quantitative hazard exposure score with literature-backed thresholds rather than a subjective five-point assessment.

Specifically, for each facility, the platform pulls the historical climate profile from National Aeronautics and Space Administration POWER and computes ten-year and thirty-year trends in temperature maxima, precipitation extremes, wind speed maxima, and dry-spell durations. Combined with the Intergovernmental Panel on Climate Change Sixth Assessment Report Shared Socioeconomic Pathway 2-4.5 and Shared Socioeconomic Pathway 5-8.5 projection data, the platform produces a quantitative climate risk profile for 2030 and 2050. The Champion's qualitative assessment is preserved (it captures local knowledge that climate models cannot) but is now triangulated against the quantitative climate record.

10.4 Adaptation recommendation engine

After computing the score, the engine generates a ranked list of recommended interventions from the intervention catalogue. For each intervention, the platform checks prerequisites, estimates the expected resilience gain using the Bayesian Adaptation Model (Section 11), estimates the cost using the cost formula in the catalogue (typically a function of facility size), and computes a ranking score as the expected gain times the clinical criticality weight divided by the cost. The top ten interventions are returned with full justification.

10.5 The Resi-Health Grid Climate Vulnerability Index

The Resi-Health Grid extends the Climate-Resilient Primary Health Care framework with a fully geospatial, externally verifiable Climate Vulnerability Index that is computable for any facility from coordinates alone. Where the original framework relied on Champion self-reporting on a five-point scale per hazard, the Resi-Health Grid produces a quantitative score grounded in three vectors drawn from open geospatial data: Hazard, Exposure, and Adaptive Capacity.

The Hazard vector measures the historical frequency and intensity of localised extreme weather at the facility's location. This is derived from the National Aeronautics and Space Administration POWER record (forty years of daily observations), the European Centre for Medium-Range Weather Forecasts Re-Analysis version 5 (sixty years of hourly reanalysis), and the Climate Hazards Group Infrared Precipitation with Station data record (five-kilometre-resolution rainfall for Africa). The engine computes return-period statistics for flood-relevant precipitation events, heatwave occurrence rates, and consecutive-dry-day frequencies.

The Exposure vector measures the facility's geographic proximity to active flood plains, high-heat zones, coastal storm tracks, and other hazard-bearing terrain. This is derived from Copernicus Emergency Management Service flood hazard layers, the Joint Research Centre Global Surface Water dataset, OpenStreetMap waterway geometries, and elevation data from the Shuttle Radar Topography Mission. Distance to the nearest flood polygon, elevation above the nearest river, and proximity to documented landslide zones are each encoded as continuous variables.

The Adaptive Capacity vector measures the facility's existing ability to absorb and respond to climate stress. Variables include current backup power configuration (solar, diesel, grid, none), battery autonomy in hours, water storage volume in litres, distance to the nearest higher-tier facility, and the Energy Performance Index from Section 9.6.

The three vectors are combined through a Bayesian hierarchical model that produces a posterior Climate Vulnerability Index in the range zero to one hundred, stratified by hazard type (flood, drought, heat, storm) and projected forward to 2030 and 2050 under both Intergovernmental Panel on Climate Change Sixth Assessment Report Shared Socioeconomic Pathway scenarios. The output appears on the Ubuntu AfyaLink Administrator dashboard as a dynamic risk map: every facility coloured by composite

Climate Vulnerability Index, with the ability to filter by hazard type and projection year, and with a high-risk priority list ordered by composite score multiplied by catchment population.

The Resi-Health Grid is consumed by the Adaptation Recommendation Engine in Section 10.4 as an additional ranking input. High-vulnerability facilities are prioritised over low-vulnerability facilities, and the financial-modelling layer recommends the appropriate Energy Conservation Measure bundle for each tier.

Part 11 — AfyaSolar Intelligence (Artificial Intelligence Subsystem)

11.1 Architectural overview

The Artificial Intelligence subsystem runs as a separate Python sub-project under source-machine-learning. It has its own dependencies (managed by uv or pdm), its own test suite, its own continuous integration workflow, and is deployable as one of three artefacts: training jobs, the serving service (a FastAPI application exposing inference endpoints), and batch inference scripts (standalone Python scripts triggered by the scheduler). The Node.js side never imports Python. The two sides communicate exclusively via Hypertext Transfer Protocol (for synchronous inference) and via BullMQ jobs (for asynchronous training and batch).

11.2 The ten model families

Model one — Daily solar generation prediction

Predicts tomorrow's solar generation given the seven-day climate forecast. Inputs: forecast irradiance, forecast cloud cover, forecast temperature, facility's installed photovoltaic capacity, facility's roof orientation and tilt. Outputs: daily kilowatt-hours expected for each of the next seven days with uncertainty bands. Training approach: pvlib produces a physical-model baseline; a gradient-boosted regression model (LightGBM) trained on residuals between pvlib prediction and manual weekly meter readings learns the facility-specific corrections (soiling factor, shading factor, panel degradation rate).

Model two — Monthly energy demand prediction

Predicts monthly energy demand for any facility given facility characteristics and climate. Inputs: facility level, patient volume, equipment inventory, climate variables for the month. Outputs: expected monthly kilowatt-hours with confidence interval. Training approach: gradient-boosted regression (XGBoost) trained on facilities for which at least three months of measurement exist. This model powers sizing recommendations for new facilities that have not yet been visited.

Model three — Facility climate-vulnerability classification

Classifies each facility into a climate-vulnerability tier (low, medium, high, extreme) for each hazard (flood, drought, heat, storm). Inputs: thirty-year climate statistics for the facility coordinates, ten-year trends, projected 2030 and 2050 values under both Shared Socioeconomic Pathway scenarios. Training approach: rule-based scoring using established hazard-classification frameworks (World Bank Climate Change Knowledge Portal, World Resources Institute Aqueduct), refined with logistic regression once a year of facility-level outcome data exists to calibrate against. This model produces the District Medical Officer's ranked-risk list.

Model four — Grid outage probability prediction

Predicts probability of grid outage in the next twenty-four to seventy-two hours at a facility. Inputs: forecast precipitation intensity, forecast wind speed, lightning probability, day-of-week, time-of-year, historical outage pattern at that location. Training approach: logistic regression initially, gradient-boosted

tree once data sufficient. Training data: outage events reported by Energy Champions, timestamp-correlated to climate at the time. This drives the Wednesday-evening pre-positioning alert.

Model five — Heatwave health alert

Detects heatwave conditions five days in advance and warns facility staff. Inputs: forecast wet-bulb globe temperature for the next five days, facility characteristics (ward layout, vaccine fridge specifications, neonatal services offered). Training approach: deterministic physical model plus thresholding, not machine learning. Thresholds from the Ahmedabad Heat Action Plan literature, adjusted for tropical-acclimated populations. No training data required beyond threshold calibration.

Model six — Climate-disease early warning

Predicts elevated risk of malaria, dengue, or cholera in the facility's catchment in the next two to four weeks. Inputs: recent rainfall pattern, temperature, humidity, historical seasonal disease pattern. Training approach: replicates published methodology (Mozambique malaria forecasting work of Colborn et al 2018, Kenya highland malaria work of Hay et al 2002, South Africa-Vietnam Temporal Fusion Transformer work). This is academic partnership work — co-authored with Ifakara Health Institute or Muhimbili University.

Model seven — Adaptation intervention ranking per facility

For each facility, ranks candidate Energy Conservation Measures and adaptation interventions by expected resilience gain per dollar. Training approach: Bayesian model averaging. Each intervention starts with a literature-based prior on its effect size. As interventions are deployed and outcomes observed (through periodic resilience re-assessment), the priors update toward empirical reality. With one hundred deployed interventions per category, priors become data-driven rather than literature-driven.

Model eight — Cold-chain risk prediction without telemetry

Predicts probability of cold-chain disruption at a facility in the next twenty-four to seventy-two hours. Inputs: forecast ambient temperature, forecast grid outage probability (Model four), refrigerator age and type, facility's resilience score, time of year. Training approach: survival analysis or gradient boosting on cold-chain failure events reported by Champions. Without telemetry inside the fridge, this model cannot detect specific equipment failure modes, but it can identify the facilities at highest contextual risk in the next seventy-two hours.

Model nine — Counterfactual baseline for impact attribution

Given a facility's pre-intervention period, predicts what its energy consumption and resilience would have been without the intervention, adjusted for climate. Training approach: linear regression with climate variables as covariates, fitted on the pre-intervention period and projected forward. This is the standard International Performance Measurement and Verification Protocol Option C methodology, properly executed. Without climate adjustment, savings claims confound intervention effects with weather variation. With climate adjustment, savings are attributable. This is the spine of every credible savings claim in the platform.

Model ten — Generative Artificial Intelligence co-pilot

A conversational interface in Swahili and English that answers facility-staff questions, grounded in the facility's specific climate profile, equipment inventory, and recent assessment. Architecture: Retrieval-

Augmented Generation with three-tier routing. Tier A (50 percent of queries): pattern match against frequently-asked-question templates, no Large Language Model required. Tier B (40 percent): Llama 3.1 Instruct 8 billion parameters, quantised to 4-bit using GPT-Generated Unified Format, served via vLLM on a single Graphics Processing Unit. Tier C (10 percent): cloud Large Language Model fallback (Claude or equivalent) with explicit user consent and prompt scrubbing of patient-level data.

Embedding model for retrieval: intfloat multilingual-e5-large (1024 dimensions). Vector store: pgvector inside the main PostgreSQL instance. Refusal patterns: explicit list at source-machine-learning-copilot-refusal-patterns.yaml, covering clinical diagnostic queries, medication dose queries, and regulatory rulings (all route to qualified humans, never answered by the model).

Model eleven — Eco-Pulse Energy Performance Index

Computes the Energy Performance Index per facility as defined in Section 9.6. Inputs: facility tier, gross floor area, bed count, outpatient volume, appliance inventory, monthly ambient-temperature profile from Open-Meteo, regional climate normals from National Aeronautics and Space Administration POWER. Training approach: a Multiple Linear Regression model is fit on the structural variables to produce an expected baseline; a Generalised Additive Model is fit on the temperature-by-load interaction to capture non-linear cooling demand; outputs are stacked through ridge regression. The model is retrained quarterly. Output: a continuous Energy Performance Index score, an anomaly flag where the score deviates from the peer-tier baseline by more than one standard deviation, and a ranked diagnostic-hypothesis list that links to the Energy Conservation Measures catalogue.

Model twelve — Synthetix-HFE Satellite Anomaly Detection

Detects data-integrity anomalies in the National Health Facility Registry through monthly Visible Infrared Imaging Radiometer Suite Day-Night Band imagery as defined in Part 12. Inputs: facility coordinates, twenty-four months of monthly radiance observations per facility, declared electrification status from the registry. Training approach: a robust z-score on the rolling twenty-four-month radiance distribution with regional baseline-noise calibration; supplemented by an isolation forest for multivariate outlier detection once seasonal patterns accumulate. Output: a flagged-facility list with anomaly type (electrification gain, electrification loss, candidate new build, candidate asset loss), confidence score, and a recommended action (trigger validation loop, escalate to Administrator, schedule technician visit).

11.3 What the AI does for each persona

For the Facility Manager

- Simulate: she can ask what if questions in WhatsApp (what if we added a second fridge, what if the rains start late). The platform simulates the implications and replies.
- Forecast: the daily six-thirty WhatsApp message contains a seven-day forecast specific to her facility.
- Predict: predictive cold-chain alerts arrive two to four weeks in advance for failing refrigerators.
- Warn: heatwave alerts, flood alerts, outage alerts arrive with sufficient lead time for action.
- Plan locally: adaptation recommendations are localised to her facility's climate, services, and budget.

For the Ubuntu AfyaLink Administrator

- Simulate: she can model portfolio-wide intervention scenarios (what would happen if we replaced all gas-absorption fridges this year).
- Forecast: portfolio-wide revenue and operating cost forecasts based on confirmed contract pipeline.
- Predict: which facilities are likely to default on payments, which technicians are over- or under-utilised, which Champions are at risk of disengaging.
- Warn: portfolio-level alerts when multiple facilities in a region face simultaneous climate stress.
- Optimise: constrained-optimisation recommendations for allocating a fixed grant across the portfolio for maximum resilience gain.

For the Impact Investor

- Simulate: what would the portfolio look like with another fifty facilities added in coastal Pwani.
- Forecast: projected impact over the funder's commitment period.
- Predict: probability of meeting agreed outcome targets given current trajectory.
- Warn: notification if portfolio metrics deviate significantly from the agreed targets.
- Verify: every claim in the impact reporting links to its supporting data and methodology.

Part 12 — The Synthetix-HFE Self-Healing Data Integrity Module

National health facility electrification databases are notoriously plagued by data drift. A facility registered as "fully electrified with grid-tied solar" three years ago may have lost its array to theft, may have a failed inverter, or may have moved to diesel-only operation, and no entry in the database will reflect this until a manual update is filed. Manual updates rarely happen at scale; ministries and donors are forced to plan against data that is months or years out of date. The Synthetix-HFE module is the platform's automated, continuous data-integrity layer. It operates a self-healing loop that does not depend on field officers or manual surveys to maintain currency.

12.1 Architecture

The module has three components: a monthly satellite audit pipeline, an anomaly detection layer, and a low-cost human-in-the-loop validation channel through Unstructured Supplementary Service Data and Short Message Service. The three components operate in a continuous cycle: the satellite pipeline pulls new imagery each month, the anomaly detection layer flags facilities whose observed radiance has drifted from their historical baseline, and the validation channel converts each flag into a confirmed status update by directly asking the facility's registered Champion.

12.2 The satellite audit pipeline

Each month the module pulls Visible Infrared Imaging Radiometer Suite Day-Night Band imagery for every facility coordinate in the National Health Facility Registry through the United States Earth Observation Group's open-data feed. Night-time light radiance at each coordinate is extracted and

stored as a time-series in a dedicated TimescaleDB hypertable. The historical radiance profile per facility is computed across the preceding twenty-four months. A facility that the registry describes as "fully electrified" but whose recent night-time radiance has dropped substantially below its historical mean is flagged as a potential data mismatch. A facility described as "unelectrified" whose night-time radiance has risen substantially is flagged as a candidate for re-classification. The module also detects new-build candidates (radiance rising at coordinates that contain no registered facility) and asset-loss candidates (radiance falling at coordinates of registered facilities).

The detection model is straightforward: a robust z-score on the rolling twenty-four-month radiance distribution, with thresholds tuned to the regional baseline noise. False-positive control is critical because every flag triggers a downstream validation cost. The model is conservative by default and is refined quarterly as ground-truth from the validation loop accumulates.

12.3 The validation loop

A flagged facility does not result in an automatic registry update. The module issues an automated, low-cost validation prompt through Africa's Talking Unstructured Supplementary Service Data or Short Message Service to the registered phone number of the facility's nurse-in-charge or designated Energy Champion. The prompt reads: "System alert: Our mapping indicates a possible change in power status at your facility. Reply 1 if you are fully powered, 2 if running on backup or partial power, 3 if facing a complete outage. Reply HELP for assistance."

Responses are returned through the Short Message Service or Unstructured Supplementary Service Data channel and are recorded as a verified ground-truth observation in the platform. If the facility confirms the satellite-indicated change, the registry is updated and the relevant Ubuntu AfyaLink Administrator and Energy Champion are notified. If the facility contradicts the satellite signal, the contradiction is logged, the satellite threshold is held, and the case is escalated to the Administrator for a follow-up call or visit.

12.4 What it produces

For Ubuntu AfyaLink, the module produces a continuously current, low-friction operational map of which facilities in the portfolio are actually functional at any given moment, independent of self-reported status.

For the Ministry of Health, it produces a continuously current national heatmap of facility electrification status, with auditable provenance for every entry (satellite observation date, validation response date, responder phone number anonymised).

For the Impact Investor, it produces independent verification that the facilities in a portfolio are operational, not derelict-on-paper.

For the platform's commercial proposition, Synthetix-HFE is the operational answer to the recurring donor question, "how do you know your facilities are still running?" The answer is the satellite, the time-series, and the validated response from the registered Champion's phone, all timestamped, all signed, all citable.

12.5 Where it lives in the codebase

Implementation lives at source-library-datatwin-synthetix. The satellite ingestion runs as a scheduled BullMQ job at the first of every month. The anomaly detection runs immediately after ingestion completes. The validation loop runs as an Africa's Talking webhook handler in the Web tier. Schema

additions: a `facility_radiance_observations` TimescaleDB hypertable storing one row per facility per month, a `facility_data_anomalies` table recording each flag with type and confidence, and a `facility_validation_responses` table recording each Champion reply against its triggering anomaly.

Part 13 — Energy-as-a-Service Pricing and Innovative Financing Mechanisms

13.1 The pricing principle

Under the re-invented Energy-as-a-Service model, the facility pays installation cost upfront and then a fixed monthly fee thereafter. This is a deliberate variation from the typical Energy-as-a-Service model (where the customer pays nothing upfront and an above-market monthly fee covers both capital recovery and operating cost) for two reasons. First, paying a portion upfront creates buy-in: the facility has skin in the game and is more likely to engage with the platform as a partner rather than as a passive customer. Second, paying a portion upfront reduces the monthly fee, which makes the contract dramatically more affordable from month one, which improves adoption. The price the facility actually cares about is the monthly fee, which is now meaningfully below the facility's current energy spend.

13.2 The pricing formula

For each facility's recommended solar system, the platform computes pricing under three financing models.

Cash model

Facility pays the full system capital expenditure upfront. Total approximately equal to the sum of: array kilowatt-peak times installed cost per kilowatt-peak, plus battery kilowatt-hours times installed cost per kilowatt-hour, plus inverter kilo-volt-amperes times installed cost per kilo-volt-ampere, plus balance of system (cables, mounting, switchgear, installation labour), plus a markup for working capital and risk. Facility takes title immediately. No further monthly fees other than optional monitoring and maintenance contract.

Installment model

Facility pays approximately fifteen percent of capital expenditure as installation deposit. Remainder amortised over twenty-four to forty-eight months at an interest rate that reflects Ubuntu AfyaLink's cost of capital plus risk margin. Facility takes title at completion of payment. Maintenance contract optional.

Energy-as-a-Service model

Facility pays the installation cost upfront (typically fifteen to twenty-five percent of total system cost). Thereafter pays a fixed monthly fee for seven years. The monthly fee covers capital recovery on the remaining seventy-five to eighty-five percent of the system cost, plus operating cost (maintenance, monitoring, technician visits, replacement parts), plus a margin. The monthly fee is structured to be lower than the facility's current energy spend, so cash flow improves from month one. At the end of seven years, asset transfers to the facility at zero or nominal cost. Maintenance contract continues optionally at a reduced rate.

13.3 Specific example for Ubuntu Facility, Kinondoni, Dar es Salaam

Using the platform's pricing engine for the 30 kilowatt system displayed in the current platform user interface:

Item	Amount (Tanzanian Shilling)	Note
System capital expenditure	36,900,000	Total cash price
Installation cost (upfront under Energy-as-a-Service)	7,380,000	20 percent of total
Amount financed under Energy-as-a-Service	29,520,000	Remaining 80 percent
Monthly Energy-as-a-Service fee	814,000	Over 7 years (84 months)
Total Energy-as-a-Service payments over 7 years	68,376,000	Including financing margin
Estimated current monthly energy spend	Approximately 1,200,000	From baseline audit
Monthly saving from month 1	Approximately 386,000	Versus current spend
Asset transfer at end of year 7	Zero	Title transfers to facility

13.4 Pricing parameters and how they connect to recommendations

The monthly Energy-as-a-Service fee is not a fixed number that the facility manager negotiates. It is computed automatically from the platform's recommendations and cost-effectiveness analysis.

Specifically, the fee is determined by: the recommended system size (which is determined by the climate-data-driven sizing engine), the recommended battery capacity (which is determined by the worst-three-day climate analysis), the cost-effectiveness threshold (the fee must produce a positive saving from month one), the term length (default seven years, adjustable), and the operator's working capital cost. The facility manager sees the result, not the calculation. She sees: this system. This installation cost. This monthly fee. This monthly saving versus current spend. This break-even date. This asset-transfer date. Each number is the output of the platform's optimisation.

13.5 The Revenue-Linked Smart-Splitter Gateway

A rigid, fixed monthly Energy-as-a-Service fee assumes that a clinic's cash flow is steady. In practice, primary healthcare facility revenue is highly variable: patient volumes spike during outbreaks and harvest seasons and trough during planting and rainy seasons. A fixed monthly fee that is comfortable in a busy month is punishing in a slow month, and the operational consequence is either a missed payment (triggering technical default) or a deferred payment (eroding the operator's working capital).

The Revenue-Linked Smart-Splitter Gateway resolves this misalignment by binding the monthly fee directly to the clinic's daily digital revenue rather than to a calendar. The result is an Energy-as-a-Service contract that automatically accommodates revenue variability without renegotiation.

13.5.1 Architecture

The Gateway is a software integration that sits between the clinic's primary digital revenue-collection point and the clinic's main operating bank account. The primary integration partner is Tigo Pesa through the Tigo Lipa merchant payment service. The Gateway is positioned at the merchant Till or Lipa number

that receives patient payments and disburses two streams: a small percentage routed to the solar developer's account and the remainder forwarded to the clinic's operating wallet.

In flow terms: a patient digital payment arrives at the Tigo Lipa number; the Gateway applies the contractual auto-split (typically two to five percent to the Ubuntu AfyaLink developer account, the remainder to the clinic main operating wallet); both legs settle the same day; both parties receive matching reconciliation receipts.

13.5.2 The flexible payment formula

Each day the Gateway computes the day's payment to the operator as a function of the clinic's gross digital receipts for that day, subject to a contractual monthly cap. The daily payment formula is: daily payment equals the lesser of (alpha times daily revenue) or (monthly cap minus cumulative paid this cycle). Where alpha is the agreed split coefficient (typically two to five percent of gross daily receipts), daily revenue is the clinic's gross digital receipts on that day, monthly cap is the contractual monthly fee ceiling, and cumulative paid is the running total already routed within the current billing cycle.

The economic effect is precise. On a high-revenue day the split percentage applied to a larger denominator catches the operator up on the monthly obligation more quickly. On a low-revenue day the same percentage produces a smaller absolute amount that the clinic can comfortably absorb. Once the contractual monthly cap is reached, the split is automatically suspended for the remainder of that billing cycle and the clinic retains one hundred percent of subsequent receipts until the next cycle begins.

13.5.3 The Tigo Lipa partnership

Ubuntu AfyaLink's commercial implementation of the Gateway is built on Tigo Pesa's Lipa merchant Application Programming Interface. The partnership provides a dedicated merchant Till number per facility branded for the clinic, an automated split rule applied at the moment of payment receipt before settlement to the clinic, real-time reconciliation of split payments with the platform's contract ledger, and statement integration so that both the clinic and Ubuntu AfyaLink receive matching daily settlement reports.

The technical integration is implemented through Tigo's Business-to-Business and Business-to-Customer Application Programming Interfaces. Webhooks fire on every transaction; the platform writes a payment record, updates the facility's contract ledger, and reconciles the cumulative monthly payment against the contractual cap.

13.5.4 De-risking effect

For the clinic, default risk effectively disappears. There is no fixed date on which a payment is missed; the daily split is invisible at the operational level and the cumulative monthly cap protects against over-payment in a high-revenue month. For Ubuntu AfyaLink, payment realisation is faster on average because the operator collects continuously rather than waiting for a single monthly settlement. For commercial lenders financing the facility's system, the revenue-linked structure produces measurable cash-flow predictability and is materially more bankable than a fixed-fee contract enforced through after-the-fact remedy.

13.6 The Insurance Claims Receivables Escrow

A significant share of any self-paying facility's revenue is tied up in outstanding claims from the National Health Insurance Fund and from private medical insurance schemes. Settlement of these claims is routinely delayed by sixty to one-hundred-and-twenty days, and the resulting cash-flow lag is the single

most common operational complaint of primary healthcare facility managers. The lag also distorts the lender's risk picture: a clinic that is technically profitable may be operationally illiquid.

The Insurance Claims Receivables Escrow is the platform's structural response to this lag. It uses a standardised legal instrument (a Notice of Assignment of Receivables) combined with an automated software protocol to route a fixed, designated portion of incoming National Health Insurance Fund payouts directly into a protected solar-payment escrow account before the residual balance is forwarded to the clinic's main operating account.

13.6.1 The Notice of Assignment

The facility signs a Notice of Assignment of Receivables as part of the Energy-as-a-Service contract. The Notice specifies the percentage of upcoming National Health Insurance Fund payments that will be diverted into the escrow account up to the contractual monthly Energy-as-a-Service fee. The Notice is co-signed by Ubuntu AfyaLink and is filed with the National Health Insurance Fund through the existing claims-administration channel. The legal effect is that the National Health Insurance Fund treats Ubuntu AfyaLink as the assignee of the relevant portion of the receivable and pays the escrow account directly.

13.6.2 The Intercept Account

The escrow account is a regulated trust account held with a Tanzanian commercial bank. The platform mirrors the account's balance, transaction history, and reconciliation status in the Ubuntu AfyaLink Administrator dashboard. When a National Health Insurance Fund payment lands in the escrow, the contractual fee for the relevant month is retained, any surplus above the contractual fee is automatically forwarded to the clinic's main operating wallet (typically within one banking day), and a reconciliation event is written to the platform's contract ledger and is visible to both Ubuntu AfyaLink and the facility manager.

13.6.3 Why this is a structural risk reduction, not a payment reshuffle

Under a conventional contract, the commercial lender or solar investor financing the facility's system underwrites the credit risk of the individual rural clinic. The lender must price for the clinic's revenue volatility, its operational management quality, and its exposure to local shocks. Under the Insurance Claims Receivables Escrow, the lender is underwriting the credit risk of the National Health Insurance Fund itself rather than the individual clinic. The National Health Insurance Fund is a sovereign-backed entity with a multi-year track record of payment; its credit profile is materially stronger than any individual clinic's.

The practical consequence is access to lower-cost capital. Local commercial banks that would not lend to a rural primary healthcare facility on its own balance sheet will lend against a structured Notice-of-Assignment-backed cash-flow because the underwriting question has changed from "will this clinic remain solvent?" to "will the National Health Insurance Fund continue to pay claims?" The reduction in perceived risk lowers the interest rate paid by Ubuntu AfyaLink, lowers the corresponding monthly Energy-as-a-Service fee paid by the facility, and removes the prohibitive collateral requirements that have historically blocked rural healthcare energy lending in Tanzania.

13.6.4 Where it lives in the codebase

Implementation lives at source-library-finance-receivables. Schema additions: a `receivables_assignments` table that records the executed Notice per facility, an `escrow_accounts` table that holds the balances mirrored from the trust account, an `escrow_inflows` table that records payments

received from the National Health Insurance Fund, and an escrow_disbursements table that records the contractual retention and the residual forwarded to the clinic.

13.7 The Regional Cluster Insurance Protocol (Decentralised O&M)

The single largest hidden cost in rural solar operations and maintenance is the isolation tax. A failed inverter in a remote mountain dispensary may require ten dollars of replacement parts and ninety minutes of competent labour, but the mobilisation cost of dispatching a developer's engineer from Dar es Salaam to that dispensary can exceed five hundred United States dollars in fuel, daily allowance, and lost time. The economic asymmetry between the fix and the mobilisation is what produces multi-week outages at facilities that could in principle have been repaired in a single afternoon.

The Regional Cluster Insurance Protocol restructures rural operations and maintenance around two principles: geographic aggregation of facilities into maintenance circles and decentralised dispatch to local certified technicians.

13.7.1 Regional Maintenance Circles

The platform aggregates geographically proximal health facilities into named Maintenance Circles. Circles are defined by drive-time radius (typically forty-five minutes by standard ground vehicle) rather than by administrative district, and they include facilities regardless of ownership type (public, faith-based, private for-profit, private not-for-profit) because the operational problem is identical across ownership categories. Each Circle has a Circle coordinator (usually one of the Energy Champions in the cluster), a roster of certified independent local technicians whose service areas overlap with the Circle, and a dedicated reserve wallet.

13.7.2 The Micro-Premium Pool

Each Maintenance Circle holds a fractional-reserve digital wallet capitalised through a small, fixed percentage of the documented energy-cost savings achieved by every facility in the Circle relative to its pre-intervention diesel and grid baseline. The percentage is typically two to four percent of the verified monthly saving; the contribution is automatic and is routed by the platform at the same monthly reconciliation cycle as the Energy-as-a-Service fee. Over years, the pool accumulates to a level sufficient to fund the major mid-life replacement events (typically year-five battery refresh, year-seven to year-eight inverter refresh) without requiring a new financing event.

The pool is not insurance in the regulatory sense; it is a shared maintenance reserve owned collectively by the facilities in the Circle and administered by Ubuntu AfyaLink as the technical operator. The structure resembles a savings and credit cooperative more than a commercial insurance product, which keeps it culturally and legally familiar in the Tanzanian context and avoids the regulatory perimeter of the Tanzania Insurance Regulatory Authority.

13.7.3 Decentralised dispatch protocol

When the platform's anomaly detection layer identifies a fault at a facility (through the Eco-Pulse Energy Performance Index anomaly flag, the Synthetix-HFE night-time radiance drop, the Champion's WhatsApp problem report, or a manual entry) the dispatch protocol does not route the alert to the operator's central office. The protocol issues a geofenced broadcast through Short Message Service and WhatsApp to the certified local technicians whose service area covers the facility's coordinates.

The broadcast contains the facility name and location, a description of the suspected fault, the estimated parts requirement (drawn from the Energy Conservation Measures catalogue), the authorised

maximum payment for the visit, and a tokenised service ticket that grants the technician temporary platform access to log the visit, upload photographs, and submit a closure report. The first qualified technician to accept the ticket is dispatched. Acceptance is binding; the platform tracks acceptance rate, response time, and closure quality across the technician roster, and persistent under-performance results in removal from the geofenced broadcast list.

13.7.4 Why this travels

The Regional Cluster Insurance Protocol is structurally country-agnostic. The legal structure (a collective maintenance reserve administered by the platform operator) is replicable in any jurisdiction with basic mobile money infrastructure and a population of competent independent electricians. The platform code base supports the model multi-tenant: a Kenyan operator running the same protocol on its facilities pays the same platform licence and operates under its own jurisdiction's regulatory perimeter without affecting the Tanzanian implementation.

The economic effect is to convert the rural operations and maintenance cost curve from an emergency cost structure dominated by mobilisation expense into a routine cost structure dominated by labour and parts. The mid-life replacement risk that would otherwise sit on the operator's balance sheet is pre-funded from the documented savings stream, which converts the inverter-replacement event from a financing problem into a reserve-drawdown event. This is precisely the structural improvement that Development Finance Institutions require before committing scale capital to a rural healthcare energy book.

Part 14 — User Experience and Interface Guidelines

14.1 Design philosophy

The platform's design discipline is governed by five principles. First, voice replaces text wherever possible because the primary user is functionally literate but not text-comfortable. Second, pictures replace words wherever possible because pictograms cross language barriers and reduce reading load. Third, one tap replaces three taps for any common action. Fourth, every interaction takes under ninety seconds to complete. Fifth, asynchronous design is mandatory because connectivity is unreliable.

14.2 Visual design system

The design system is built on Tailwind CSS 4 with custom design tokens defined in `source-styles-tokens.css`. The brand colour is a deep forest green (oklch zero point fifty-five zero point eighteen two-twenty) chosen to differentiate from competing platforms that use blue or orange. The type scale is modular (ratio 1.250) starting at 0.75 rem extra small and rising to 4 rem display. Spacing uses a 4-pixel base scale. Shadows are used sparingly to maintain a flat, performance-oriented aesthetic that renders quickly on low-end devices.

14.3 Accessibility

Target compliance is Web Content Accessibility Guidelines version 2.2 Level AA. Specific requirements include accessible name on every interactive element, colour contrast minimum 4.5 to 1 for normal text, focus indicators visible on every focusable element, keyboard navigability throughout, programmatic label association for form fields, screen-reader-announced error messages, tap targets minimum 44 by 44 pixels, meaningful alt text on all images. Automated accessibility testing uses axe-core in playwright end-to-end tests; any violation in a non-experimental route blocks the build.

14.4 Internationalisation

Internationalisation uses next-intl. Two locales: Swahili (default) and English. Message catalogues at `public-locales-sw.json` and `public-locales-en.json`. Catalogues are organised by feature path matching the route structure. Every user-facing string lives in the catalogue; the linter rejects literal strings in JavaScript Extensible Markup Language. Swahili messages are written by a native Swahili speaker (not machine translated). Two registers are maintained: `kawaida` (everyday) for in-application text, and `rasmi` (formal) for generated reports.

Part 15 — WhatsApp, Short Message Service, and Unstructured Supplementary Service Data Channels

15.1 Channel architecture

Three channels do different work. The Progressive Web Application is the deep-work surface. WhatsApp is the daily-touch surface. Short Message Service is the floor. The implementation pattern: every notification the platform sends is classified by criticality. Informational notifications go to WhatsApp only. Important notifications go to WhatsApp with Short Message Service backup. Critical notifications go to WhatsApp and Short Message Service simultaneously, with voice call escalation if neither confirms within thirty minutes.

15.2 WhatsApp conversational state machine

Conversation state per Energy Champion is stored in a `whatsapp_conversations` table keyed by facility identifier and user identifier. The state machine has eight states: idle, awaiting one-time password, in troubleshoot fridge, in report problem, in daily report, in voice query, awaiting consent, in onboarding. Transitions are deterministic given current state and incoming message kind (text, button, list, voice, image, location). The state machine is unit-tested exhaustively.

15.3 The daily six-thirty WhatsApp push

Implementation: a BullMQ repeatable job runs hourly. For each facility scheduled for a six-thirty local push, the job confirms it is between six-thirty and six-thirty-five in that facility's timezone, reads the latest fridge temperature, battery State of Charge, expected solar today, and any pending tasks, composes a Swahili message using a template from `public-locales-sw.json`, sends through the WhatsApp Cloud Application Programming Interface as a template message, and records the send with delivery status.

15.4 Voice query handling

When the Champion sends a voice note, the webhook receives a `messages.audio` event with a media identifier. Pipeline: download the audio from WhatsApp Media Application Programming Interface, upload to Simple Storage Service for retention and analysis, transcribe using Whisper-large-v3 with Swahili language hint, pass transcription to the Generative Artificial Intelligence co-pilot, send response back. End-to-end latency target: less than eight seconds from voice note received to response sent.

15.5 Short Message Service and Unstructured Supplementary Service Data fallback

Where WhatsApp is not available, the same conversational state machine is exposed through Short Message Service and Unstructured Supplementary Service Data through Africa's Talking. For Unstructured Supplementary Service Data, the menu is shallower: dial code 123*456 hash returns a menu with options for fridge status, power status, problem reporting, and daily reporting. For Short Message Service, syntax

is keyword-based: STATUS returns a one-message status summary; REPORT 45 12 2 submits a daily report (45 patients, 12 children vaccinated, 2 deliveries); HELP returns supported keywords.

Part 16 — Implementation Roadmap

16.1 Phase one (Week one through two) — Foundation

- Climate data ingestion pipeline operational: NASA POWER daily pull, ERA5 monthly pull, CHIRPS monthly pull
- Database migration from MySQL to PostgreSQL with TimescaleDB and pgvector
- Repository restructure to support Python machine learning sub-project
- Climate observations table populated for all existing thirteen facilities
- Renaming: Climate-Resilient Primary Health Care Operating System as the platform, AfyaSolar as the reference implementation
- Massachusetts Institute of Technology licence applied to the repository
- STANDARDS.md master conformance document published
- Methodology paper version 1 drafted and shared with one academic partner for co-authorship review
- Eco-Pulse Engine version 1 deployed: Multiple Linear Regression and Generalised Additive Model trained on the existing thirteen facilities plus open climate data; Energy Performance Index exposed on the Administrator Facilities page.
- Synthetix-HFE satellite ingestion pipeline operational: monthly Visible Infrared Imaging Radiometer Suite Day-Night Band pulls for all registered facilities; baseline radiance history populated.

16.2 Phase two (Week three through five) — Intelligence layer

- pvlib-based solar sizing engine operational, replacing existing sizing logic
- Synthetic load profile generator operational
- Climate vulnerability classifier operational
- Counterfactual baseline model operational
- Heatwave alert system operational
- First version of Generative Artificial Intelligence co-pilot operational in Swahili and English
- Three persona dashboards (Facility Manager, Administrator, Impact Investor) operational
- Energy Champion programme curriculum complete, certification process operational, first cohort enrolled
- Resi-Health Grid Climate Vulnerability Index operational: Hazard, Exposure, and Adaptive Capacity vectors combined into a Bayesian posterior; dynamic risk map live on the Administrator dashboard.
- Synthetix-HFE anomaly detection plus USSD/SMS validation loop in production; first reconciled registry corrections complete.

16.3 Phase three (week six through nine) — Scale

- WhatsApp Cloud Application Programming Interface in production replacing Twilio wrapper
- Short Message Service through Africa's Talking in production
- Mobile money integration through ClickPesa or Selcom in production
- District Health Information System version 2 integration bidirectional
- Fast Healthcare Interoperability Resources Release 5 endpoints exposed
- Digital Monitoring Reporting and Verification bundle generation pipeline operational
- First fifty facilities onboarded under the re-invented platform
- First parametric insurance partnership conversation initiated
- Tanzania Ministry of Health partnership Memorandum of Understanding signed
- Revenue-Linked Smart-Splitter Gateway live with Tigo Pesa Lipa as primary mobile-money channel; first ten facilities collecting daily auto-split payments.
- Insurance Claims Receivables Escrow operational with the National Health Insurance Fund Notice of Assignment filed for the first cohort of empanelled facilities.
- Regional Cluster Insurance Protocol launched: first Maintenance Circle formed in Pwani or Morogoro; Micro-Premium Pool capitalising from verified savings; decentralised technician dispatch operational.

16.4 Phase four (Week nine through ten) — Multi-tenant

- Multi-tenant architecture in production with first ministry-tier tenant
- Klarna AI for Climate Resilience grant deployment complete
- ASCENT Programme partnership operational with Tanzania, Kenya, Uganda regional pilots
- Eco-Pulse and Resi-Health Grid offered as standalone Application Programming Interface products to Ministries of Health and Development Finance Institutions in Tanzania, Kenya, Uganda.
- Synthetix-HFE published as a public dashboard for the Tanzania Ministry of Health and the regional Sustainable Energy for All Powering Healthcare Hub.
- Second operator onboarded onto Revenue-Linked Smart-Splitter and Regional Cluster Insurance Protocol, proving the operator-as-a-service model end-to-end.

Appendix A — Glossary and References

Glossary

Term	Definition
AfyaSolar Intelligence Platform	The technical platform operated by Ubuntu AfyaLink
Asset	Energy-consuming or generating device at a facility
Assessment	Structured point-in-time facility evaluation
Champion	Designated Energy Champion at a facility, paid stipend role
Climate-Resilient Primary Health Care	The scoring framework underlying the platform's resilience assessment
Cost per service-hour	The platform's signature metric
Digital Monitoring Reporting and Verification	Signed verifiable outcome bundle
Energy Conservation Measure	Specific intervention reducing consumption or improving efficiency
Energy-as-a-Service	Financing model with installation cost upfront plus fixed monthly fee
Facility	Primary healthcare facility — dispensary, health centre, polyclinic, or hospital
International Performance Measurement and Verification Protocol	International standard for verifying energy savings
Plan	Costed sequenced intervention set with expected impact
Service-hour	One hour with cold chain functional, lighting available, critical equipment powered
Tenant	Logical owner of facility data — operator, ministry, or network
Eco-Pulse Engine	Software-only energy efficiency benchmarking layer combining Multiple Linear Regression and Generalised Additive Models to produce the Energy Performance Index per facility
Energy Performance Index (EPI)	Ratio of observed to expected monthly energy consumption for a facility, adjusted for climate; one is the peer-baseline expectation
Resi-Health Grid	Geospatial climate-vulnerability mapping engine combining Hazard, Exposure, and Adaptive Capacity vectors into a Bayesian Climate Vulnerability Index
Climate Vulnerability Index (CVI)	Composite zero-to-one-hundred score produced by the Resi-Health Grid for any facility from coordinates alone
Synthetic-HFE	Self-healing data-integrity module using monthly satellite night-time-lights observations cross-validated by Champion USSD/SMS responses

Revenue-Linked Smart-Splitter	Payment gateway that auto-splits a small, formula-driven percentage of a clinic's daily mobile-money receipts to the solar developer, integrated via Tigo Pesa Lipa
Notice of Assignment of Receivables	Legal instrument by which a facility assigns a portion of its NHIF payouts directly into the Insurance Claims Receivables Escrow before reaching its operating account
Maintenance Circle	Geographic cluster of nearby health facilities (regardless of ownership) sharing a Micro-Premium Pool and decentralised technician dispatch under the Regional Cluster Insurance Protocol
Micro-Premium Pool	Shared maintenance reserve wallet per Maintenance Circle, capitalised from a small fixed percentage of each facility's verified energy-cost savings

Key references

- World Health Organization Operational Framework for Building Climate Resilient Health Systems (2015)
- World Health Organization Guidance for Climate Resilient and Environmentally Sustainable Health Care Facilities (2020)
- World Health Organization Ethics and Governance of Artificial Intelligence for Health (2021)
- American Society of Heating, Refrigerating and Air-Conditioning Engineers Standard 211-2018
- International Organization for Standardization 50001:2018 Energy Management Systems
- International Organization for Standardization 50002:2014 Energy Audits
- International Organization for Standardization 50006:2014 Energy Baselines and Energy Performance Indicators
- International Organization for Standardization 14091:2021 Climate Change Adaptation Vulnerability and Risk Assessment
- International Organization for Standardization and International Electrotechnical Commission 42001:2023 Artificial Intelligence Management System
- International Performance Measurement and Verification Protocol Core Concepts 2022
- United States National Institute of Standards and Technology Artificial Intelligence Risk Management Framework 1.0
- Intergovernmental Panel on Climate Change Sixth Assessment Report Working Group II Africa Chapter
- Tanzania Data Protection Act 2022
- Tanzania National Adaptation Plan
- Tanzania Health Sector Strategic Plan version 5
- Web Content Accessibility Guidelines version 2.2
- Digital Public Goods Standard version 1.1
- World Wide Web Consortium Verifiable Credentials Data Model 1.1

Appendix B — Document Closing

This document is the engineering source of truth for the AfyaSolar Intelligence Platform under operation by Ubuntu AfyaLink. Changes to the document are made via pull request with the label meta-engineering-spec. Major version bumps require Technical Steering Committee approval once that body is formed.

The platform's commercial proposition rests on three structural facts. First, the data substrate (climate data plus facility surveys) is genuinely sovereign and genuinely free at the marginal cost. Second, the artificial intelligence layer turns that substrate into operationally useful predictions, simulations, and recommendations at a quality that no competing platform currently matches in the African primary healthcare context. Third, the standards conformance trail (American Society of Heating, Refrigerating and Air-Conditioning Engineers Standard 211, International Organization for Standardization 50001, International Organization for Standardization 14091, International Organization for Standardization and International Electrotechnical Commission 42001, World Health Organization Guidance, Intergovernmental Panel on Climate Change Sixth Assessment Report) makes every platform output citable, adoptable, and procurable by ministries and Development Finance Institutions.

The engineering work over the next twelve months is approximately one senior software engineer at full time, one machine learning engineer at full time, and one design and front-end engineer at part time, supported by Ubuntu AfyaLink's existing leadership team (Anodi, Idda, Albert). The deliverables that the team should ship in this twelve-month period are enumerated in Part Fifteen. The platform is positioned to be the climate-resilient primary healthcare operating system of choice for East Africa by year three, scaling to broader Sub-Saharan Africa by year five.

Version 3.1 (this revision) adds three software-side intelligence modules — the Eco-Pulse Virtual Energy Performance Index engine, the Resi-Health Grid Climate Vulnerability Index, and the Synthetix-HFE Self-Healing Data Integrity Module — and three financing-and-operations innovations — the Revenue-Linked Smart-Splitter Gateway integrated with Tigo Pesa Lipa, the Insurance Claims Receivables Escrow integrated with the National Health Insurance Fund, and the Regional Cluster Insurance Protocol for decentralised operations and maintenance. These six additions extend the platform from a climate-data-substrate operating system into a complete software-defined infrastructure layer for healthcare facility electrification, with financing and maintenance mechanisms that are independently bankable and operationally country-replicable.

End of document.

Revenue-Linked Smart-Splitter

We will compare this model against a **Traditional Fixed Lease** across two operational extremes at a growth-stage health clinic: a high-revenue month (peak treatment season) and a low-revenue month (dry season/low patient traffic).

The Mathematical Framework

The automated split-payment engine calculates the daily transfer to the developer using a conditional minimization function to ensure payments never exceed a defined ceiling:

$$SP_{\text{daily}} = \min(\alpha \cdot R_{\text{daily}}, M_{\text{cap}} - M_{\text{paid}})$$

Where:

- $\$P_{\text{daily}}\$$ = The actual dollar amount routed to the solar developer on a given day.
- $\$R_{\text{daily}}\$$ = The gross digital revenue collected by the clinic via its billing systems or mobile money wallets over a 24-hour cycle.
- $\alpha\$$ = The **Split Coefficient** (the fixed, contractually agreed percentage of daily revenue dedicated to energy).
- $\$M_{\text{cap}}\$$ = The **Monthly Payment Cap**, representing the maximum amount the developer can collect in a single month (allowing them to capture upside up to a threshold).
- $\$M_{\text{paid}}\$$ = The cumulative total of all daily payments already transferred to the developer within the current calendar month ($\$M_{\text{paid}} = \sum P_{\text{daily}}\$$).

Simulation Parameters

For this simulation, we will establish standard commercial baselines:

- **Solar Asset Value:** A 10kW hybrid solar array with lithium storage.
- **Traditional Alternative:** A rigid fixed lease of **\$1,000 per month**. Falling short by even \$1 triggers a contract breach.
- **Smart-Splitter Setup:** Split Coefficient (α) set to **3%** ($\$0.03$); Monthly Cap ($\M_{cap}) set to **\$1,200**.

Scenario A: The High-Revenue Month (Peak Season)

During this period, agricultural harvests provide communities with disposable income, and seasonal disease spikes increase clinic traffic.

- **Average Daily Clinic Revenue ($\$R_{\text{daily}}\$$):** \$1,500
- **Total Monthly Gross Revenue:** \$45,000

Step-by-Step Execution:

- **Days 1 through 26:** The software runs the formula daily.

$$\alpha \cdot R_{\text{daily}} = 0.03 \cdot 1500 = \$45$$

Because $\$M_{\text{paid}}\$$ has not neared the \$1,200 cap, the system transfers **\$45 per day**. By the end of Day 26, the cumulative payment ($\$M_{\text{paid}}\$$) reaches:

$$26 \cdot \$45 = \$1,170$$

- **Day 27 Execution:** The engine checks the remaining headroom before executing:

$$\$M_{\text{cap}} - M_{\text{paid}} = \$1,200 - \$1,170 = \$30$$

The system compares the two values: $\min(\$45, \$30) = \$30$. The gateway routes **\$30** to the developer on Day 27, bringing $\$M_{\text{paid}}\$$ to exactly **\$1,200**.

- **Days 28 through 30:** Because $\$M_{\text{cap}} - M_{\text{paid}} = 0$, the formula resolves to $\min(\$45, \$0) = \$0$. The split stops automatically.

Scenario A Outcome:

The developer secures **\$1,200** (a **20% bonus** over the traditional \$1,000 lease baseline). The clinic easily retains \$43,800 of its gross revenue to finance expanded medical operations.

Scenario B: The Low-Revenue Month (The Cash Crunch)

A sudden regional economic dip or a dry spell drops patient volume significantly. The clinic struggles to cover variable operational costs like pharmaceuticals and staff salaries.

- **Average Daily Clinic Revenue (R_{daily}):** \$800
- **Total Monthly Gross Revenue:** \$24,000

Step-by-Step Execution:

- **Days 1 through 30:** The algorithm processes the lower daily intake continuously:

$$\alpha \cdot R_{\text{daily}} = 0.03 \cdot \$800 = \$24$$

Because the cumulative total never approaches the \$1,200 cap, the system transfers **\$24 every single day** without interruption.

Scenario B Outcome:

- **Total Collected by Developer:** $30 \cdot \$24 = \720 .
- **The Deferred Balance:** $\$1,200 - \$720 = \$480$ is carried forward to the next high-revenue cycle as a contractually compliant deferred balance.

Traditional vs. Smart-Splitter Performance Matrix

Metric	Traditional Fixed Lease	Revenue-Linked Smart-Splitter
High-Month Payout	\$1,000	\$1,200 (<i>Cap achieved</i>)
Low-Month Payout	\$1,000 (<i>Demanded</i>)	\$720 (<i>Collected seamlessly</i>)
Clinic Cash Strain (Low Month)	Severe. Forcing a flat \$1,000 payment out of reduced net margins threatens clinic closure.	None. The payment naturally contracts relative to daily business realities.
Contractual Status (Low Month)	✗ TECHNICAL DEFAULT. The clinic misses the flat lease target, halting the project.	COMPLIANT. The contract allows variable pacing; the balance safely defers.
Developer Collection Costs	High (Requires legal notices, phone calls, or asset repossession threats).	Zero (Automated software splits funds at the point of transaction).

The Proof: How This Architecture Eliminates Default Risk

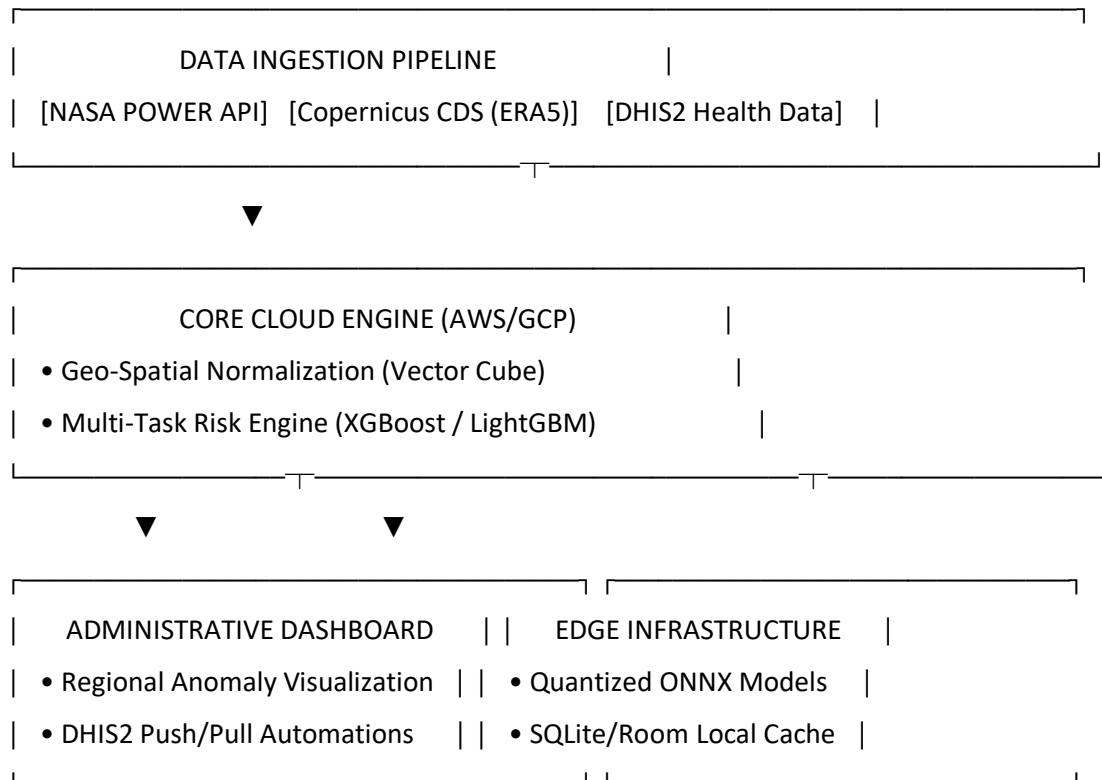
The architecture prevents defaults by shifting the relationship from an adversarial fixed debt structure to a collaborative, equity-like cash flow alignment:

1. **Systemic Volatility Absorption:** In a traditional arrangement, a drop in revenue forces the clinic to choose between paying for solar electricity or purchasing critical medical supplies. The Smart-Splitter transforms energy from a rigid fixed expense into a variable operating cost. The facility cannot default because the contract defines a revenue dip as a temporary adjustment in payment pace rather than a breach of terms.
2. **Predictable Portfolio Underwriting:** While the developer receives less money during a slow month (\$720 vs \$1,000), they completely avoid the steep legal fees, administrative friction, and total write-off risks associated with standard debt collection. The uncollected \$480 is not lost; it remains securely on the digital ledger as an enforceable account receivable backed by the clinic's future high-revenue months.
3. **Automated Risk Mitigation:** By capturing an extra **\$200** during peak operational months, the developer builds a structural cash buffer directly within their account. This buffer offsets lower payments during slower operational periods, keeping the broader portfolio stable.

Resi-Health Grid

1. System Architecture & Component Breakdown

The architecture is divided into three distinct layers: an automated cloud ingestion pipeline, an administrative planning portal, and a hyper-lightweight, offline-first mobile edge application.



Component A: The Cloud Core & Data Pipeline

The backend operates as an automated, serverless data pipeline that cross-references atmospheric environmental anomalies with historical pediatric health outcomes.

- **Ingestion Workers:** Chron tasks built on Python/Celery pull daily environmental variables. It extracts surface temperature, relative humidity, and precipitation from the **NASA POWER API**, alongside soil moisture and total precipitation anomalies from the **ECMWF ERA5 reanalysis dataset** via the Copernicus Climate Data Store (CDS) API.
- **The Spatial Vector Cube:** Raw data is downscaled and mapped onto a standardized geospatial grid (typically $0.1^\circ \times 0.1^\circ$ resolution, roughly 11 km) using open-source libraries like Xarray and GDAL.
- **Predictive Risk Core:** A machine learning model (such as a gradient-boosted tree framework like LightGBM) processes these environmental data streams alongside historical health data to predict pediatric anomaly surges up to 21 days in advance.

Component B: The Administrative Dashboard

Designed for district medical officers and ministry planners, this web-based console (built on React and D3.js) visualizes upcoming macro-risks.

- **Epidemiological Forecasting:** Displays a geographical heatmap highlighting districts with an elevated risk of climate-sensitive pediatric disease outbreaks (e.g., post-flood diarrheal spikes or heatwave-induced dehydration).
- **Resource Supply-Chain Automation:** If a specific district crosses a high-risk probability threshold, the platform generates automated supply-chain recommendations, calculating the exact volumes of oral rehydration salts (ORS), zinc supplements, or antibiotics needed for pre-positioning.

Component C: The Frontline Mobile App

A cross-platform mobile application (built using Flutter) optimized for low-tier, \$100 Android smartphones used by community health workers.

- **Local SQLite Cache:** Stores local patient data and diagnostic history using a local encrypted database, syncing back to the cloud only when a cellular connection is detected.
- **Quantized Diagnostic Engine:** Runs predictive machine learning models completely offline using **ONNX Runtime Mobile**, allowing on-the-ground triage without internet access.

2. Dynamic Operational Workflow

The system translates global climate data into hyper-local health actions through an automated, end-to-end processing loop.

1. Daily Spatial Ingestion: Cloud Core.

Automated workers pull climate metrics from NASA POWER and ERA5 APIs based on regional bounding box coordinates, normalizing the variables into a unified geospatial data structure.

2. Predictive Risk Calculation: Machine Learning Layer.

The cloud engine runs the predictive risk model to calculate regional threat scores using the fundamental vulnerability formula:

$$Risk = Hazard_{\{Climate\}} \times Exposure_{\{Pediatric\}} \times Vulnerability_{\{Infrastructure\}}$$

3.Upstream Supply Chain Alerts:Admin Dashboard.

Planners receive targeted alerts to dispatch pediatric medical supplies to local clinics three weeks before the predicted environmental anomaly peak.

4.Delta Ingestion & Sync:Edge Handshake.

When a frontline worker's mobile device briefly connects to a network, it downloads a lightweight delta packet containing updated 14-day localized risk vectors, while uploading anonymized triage logs.

5.Offline Edge Diagnostic Triage:Field App Execution.

Deep in the field with zero network access, the health worker inputs basic clinical signs. The localized ONNX model combines these symptoms with the cached environmental risk vector to flag high-risk cases.

3. End-to-End Integration Matrix

To scale without high ongoing operational costs, the architecture integrates directly with existing open-source data frameworks and health networks.

Integration Target	Data Protocol / Method	Operational Purpose
NASA POWER API	REST API via JSON payloads	Pulls daily solar radiation, surface temperatures, and ambient humidity metrics.
Copernicus CDS API	Python cdsapi client library	Extracts ERA5 high-resolution soil moisture and precipitation trends to assess vector breeding risks.
DHIS2 (District Health Information Software 2)	Web API (Tracker & Aggregate Capture)	Crucial for UNICEF alignment. Pulls historical pediatric admission numbers to train models and pushes early warning scores back into national health registries.
ONNX Runtime Mobile / TFLite	Native wrapper C++ for Flutter	Runs compressed, quantized machine learning classification matrices directly on device CPUs without drawing heavy battery power.

Open Standards & Interoperability: The entire application code uses the **Apache 2.0 License**, ensuring it remains free from proprietary lock-in. By adopting the **Fast Healthcare Interoperability Resources (FHIR)** data format and native **DHIS2 API integrations**, the platform functions as an adaptable plug-and-play module capable of deploying seamlessly alongside existing national health information architectures.

To ensure both maximum performance in the cloud and zero-connectivity execution at the point of care, our architecture decouples high-throughput training from hyper-local inference using a hybrid machine learning pipeline:

Cloud Infrastructure: High-Velocity Modeling with LightGBM

For macro-level risk forecasting in the cloud, we employ **LightGBM (Light Gradient Boosting Machine)**. This framework is explicitly chosen for its gradient-based one-side sampling and exclusive feature bundling, which allows it to process multi-terabyte, multi-modal geospatial data arrays (including NASA POWER and ERA5 time-series metrics) up to **10 times faster** and with a significantly smaller memory footprint than traditional deep neural networks or standard gradient-boosting algorithms. This structural efficiency drastically minimizes recurring cloud compute expenditures—allowing more grant funding to directly support community implementation—while enabling rapid, automated daily model retraining as new global climate registries publish.

Edge Infrastructure: Quantized ONNX Models for Low-Resource Environments

To translate these cloud-optimized predictive structures into fields with zero internet or mobile network access, the trained LightGBM models are exported into the interoperable **ONNX (Open Neural Network Exchange)** format and undergo **INT8 dynamic quantization**.

By converting the model’s weights from 32-bit floating-point (FP32) structures into compact 8-bit integers (INT8), we compress the local application footprint by up to **4×** and achieve over a **40× execution speedup** directly on the CPUs of commodity, \$100 Android smartphones. This optimization removes any requirement for expensive on-device GPU acceleration, minimizes battery and thermal strain on field devices, and ensures that community health workers can deliver instant, data-validated pediatric triage decisions entirely offline, directly aligning with UNICEF’s mandate for resilient Digital Public Goods.